

The polynomial and logarithmic order of dense sign graph matrices

A self-contained proof of the fixed-shape norm classification,
with an expository guide

Graph Matrices research project*

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Status of this manuscript. This is a research manuscript. The fixed-shape classification in Theorem 1.3, its expected-norm consequence, and the nonnegative half-integer logarithmic exponent have been formalized in Lean and checked by Comparator, Lean kernel replay, and independent NanoDa replay. Theorem 1.4 and the growing-moment, upper-tail, almost-sure, and asymptotic limited-independence extensions are outside this certified scope. The underlying arguments have received internal mathematical review; external human review is not claimed. No publication-priority or independent-discovery claim is made. This version expands the exposition of the proof: it adds an introduction with guiding examples, a section-by-section overview of the argument, worked examples inside the technical sections, commentary on the role of each hypothesis, and a notation index. The theorems, the lemmas, and the substance of their proofs are those of the technical report; see Appendix A for the dependency record.

Abstract

For a fixed finite simple graph with two ordered boundary sets, consider the matrix obtained by summing products of independent symmetric Rademacher edge variables over globally injective embeddings. We prove a two-sided classification of its expected operator norm, and of every fixed positive L^p size of that norm. The polynomial exponent is half the number of vertices minus the minimum separator size, with the usual isolated-interior correction. The logarithmic exponent is half the maximum, over minimum separators, of the number of components that miss both boundaries but remain attached to the separator. The boundaries may overlap, be unequal, or be empty, and the shape may be disconnected.

The upper proof uses finite Walsh occupation operators. A connected creation commutator contributes only the square root of the occupation grade; critical transitions factor exactly at their separator. The lower proof combines conditional moment concentration for connected fields, simultaneous root-label excursions, a normalized-column sign symmetry, and a contractive Fourier projection from the uncolored matrix to a prescribed role coloring. We also prove logarithmic upper-moment bounds and an upper-only limited-independence extension. All probabilistic and combinatorial estimates needed by this proof are established below, including the re-proved classical inequalities. Xu–Fu–Qiao–Tao give another proof of the same dense-sign expected-norm classification [4, Theorem 1.3].

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1 Introduction

1.1 The injective sign model

Let $H = (\mathcal{V}, E)$ be a fixed finite simple graph, with distinguished vertex roles, $h = |\mathcal{V}|$, and $e = |E|$. A *shape* also specifies two ordered lists U, V of roles, with no repetition within either list. The lists can overlap. We use the same letters for their underlying sets when taking intersections or deleting vertices.

For $1 \leq i < j \leq n$, let W_{ij} be independent uniform signs. Set $W_{ji} = W_{ij}$ and $W_{ii} = 0$. The rows of $M_{H,n}$ are indexed by injective assignments $a : U \hookrightarrow [n]$, and its columns by injective assignments $b : V \hookrightarrow [n]$. Define

$$M_{H,n}[a, b] = \sum_{\substack{\phi: \mathcal{V} \hookrightarrow [n] \\ \phi|_U = a, \phi|_V = b}} \prod_{xy \in E} W_{\phi(x)\phi(y)}. \quad (1.1)$$

In words: an entry of $M_{H,n}$ is a sum over all ways of placing the roles of H at distinct labels in $[n]$, compatibly with the row and column assignments, of the product of the signs along the edges. An incompatible pair of boundary assignments gives entry zero. The empty assignment indexes a one-dimensional space. In particular, an empty shape gives the scalar matrix $[1]$. We always use the Euclidean operator norm. All asymptotics are as $n \rightarrow \infty$ with H, U, V fixed.

The model is globally injective: labels assigned to different roles must differ, including across the two boundaries and across different components. Since the graph is simple, each monomial in (1.1) contains exactly e distinct primitive signs. These two facts are used constantly: the first is what the role coloring of Section 3 enforces automatically, and the second is what makes a monomial's sign support determine its embedding.

Two examples to keep in mind. The two smallest nontrivial shapes already display the two kinds of growth that the main theorem separates.

- (a) Let H be a single edge uv with $U = (u)$ and $V = (v)$. Then $M_{H,n}[a, b] = W_{ab}$ for $a \neq b$ and $M_{H,n}[a, a] = 0$: the matrix is the symmetric sign matrix itself, with zero diagonal. Its expected norm is of order $n^{1/2}$, as for any Wigner matrix.
- (b) Let H be the same edge, now written rx , with $U = V = (r)$. Then $M_{H,n}[a, b] = 0$ for $a \neq b$ and

$$M_{H,n}[a, a] = \sum_{j \neq a} W_{aj}.$$

The matrix is diagonal, and its norm is the largest absolute row sum of the sign matrix. Each row sum has size of order $n^{1/2}$, but the maximum of n such sums is of order $(n \log n)^{1/2}$.

Both shapes have two vertices and one edge, and in both the polynomial order is $n^{1/2}$. The difference is that in (b) the norm is a maximum over n labels of a scalar sum, and a maximum over n nearly independent quantities of size $n^{1/2}$ costs an extra factor $(\log n)^{1/2}$. In (a) no such maximum is forced: the norm of a Wigner matrix is not attained on a coordinate vector. The main theorem says exactly which shapes force such a maximum, and how many independent factors of $(\log n)^{1/2}$ they force. The mechanism is a separator: in (b) the role r is a separator that both boundaries pass through, and the vertex x is left dangling from it, touching no boundary.

1.2 Separators, floating components, and the exponents

Definition 1.1 (Separators and attached floating components). A set $S \subseteq \mathcal{V}$ is a *U-to-V vertex separator* if it meets every path from U to V . Boundary vertices may belong to S . A vertex

in $U \cap V$ supplies a path of length zero, so every separator contains $U \cap V$. Equivalently, no component of $H - S$ meets both boundary sets.

Let s be the smallest separator cardinality, and let $\mathcal{S}_{\min}$ be the family of separators of cardinality s . For $S \in \mathcal{S}_{\min}$, define

$$b_{\text{att}}(S) = \#\{C \text{ component of } H - S : C \cap (U \cup V) = \emptyset, E(C, S) \neq \emptyset\}, \quad b_* = \max_{S \in \mathcal{S}_{\min}} b_{\text{att}}(S). \quad (1.2)$$

Here $E(C, S)$ is the set of edges with one endpoint in each set.

Let i_0 count the isolated vertices outside $U \cup V$. Define

$$f = \frac{h + i_0 - s}{2}, \quad g = \frac{b_*}{2}, \quad T_{H,n} = n^f (\log(2n))^g. \quad (1.3)$$

The minimum-separator family is never empty: U itself is a separator. These definitions apply without change to empty boundaries.

Reading the definition. A component of $H - S$ that meets neither boundary is called *floating* at S ; it is *attached* if some edge joins it to S . The quantity $b_{\text{att}}(S)$ counts the attached floating components at S , and b_* is the largest such count over minimum separators. Three points deserve emphasis.

- Only *minimum* separators count. A larger separator may leave more floating components, but its contribution to the norm is smaller by a power of n , which no power of $\log n$ can compensate (Section 11).
- Only *attached* components count. A component of $H - S$ with no edge to S was already a component of H missing both boundaries. Such a component contributes a single scalar factor, not indexed by any separator label, and Section 9 shows that it carries no logarithm in expectation.
- The vertex count of a floating component affects only the polynomial exponent; each attached floating component contributes exactly one factor $(\log n)^{1/2}$, regardless of its size or of its number of attachments.

Example 1.2 (Computing the exponents). The following table lists, for a few small shapes, the quantities entering (1.3) and the resulting order $T_{H,n}$. Roles not in $U \cup V$ are interior. The first two rows are the examples (a) and (b) above.

Shape	U	V	minimum separators	h	s	i_0	b_*	$T_{H,n}$
edge uv	(u)	(v)	$\{u\}, \{v\}$	2	1	0	0	$n^{1/2}$
edge rx	(r)	(r)	$\{r\}$	2	1	0	1	$(n \log n)^{1/2}$
path $a x b$	(a)	(b)	$\{a\}, \{x\}, \{b\}$	3	1	0	0	n
path $a x b$	(a, b)	(a, b)	$\{a, b\}$	3	2	0	1	$(n \log n)^{1/2}$
edge uv , isolated z	(u)	(v)	$\{u\}, \{v\}$	3	1	1	0	$n^{3/2}$
edges uv and xy	(u)	(v)	$\{u\}, \{v\}$	4	1	0	0	$n^{3/2}$
edge uv	$()$	$()$	$\emptyset$	2	0	0	0	n

In the fourth row every separator must contain $U \cap V = \{a, b\}$, so $S = \{a, b\}$ is the unique minimum separator, and the interior role x is an attached floating component. The matrix is diagonal in the pair (a, b) with entries $\sum_x W_{ax} W_{xb}$, and its norm is the maximum of about n^2 sums of about n independent signs, of order $(n \log n)^{1/2}$. In the sixth row the component $\{x, y\}$ is floating at $\{u\}$ but not attached; the entry $M[a, b]$ is W_{ab} times the sum of W_{xy} over ordered

pairs of labels outside $\{a, b\}$, and that scalar has order n without a logarithm. In the last row both boundaries are empty, the matrix is the scalar $2 \sum_{i < j} W_{ij}$, and the empty set is the unique minimum separator.

1.3 Main results

Theorem 1.3 (Fixed-shape classification). *For every shape specified above and every fixed $p > 0$, there are constants $0 < c_{H,p} \leq C_{H,p} < \infty$ such that, for all sufficiently large n ,*

$$c_{H,p} T_{H,n} \leq (\mathbb{E} \|M_{H,n}\|^p)^{1/p} \leq C_{H,p} T_{H,n}. \quad (1.4)$$

In particular,

$$\mathbb{E} \|M_{H,n}\| = \Theta_H \left(n^{(h+i_0-s)/2} (\log n)^{b_*/2} \right).$$

The expected-norm conclusion is also proved by Xu–Fu–Qiao–Tao [4, Theorem 1.3] for the same globally injective simple sign model, including overlapping or empty boundaries. Their notation (v, h, s, a_*) corresponds to (h, i_0, s, b_*) here. This credits another proof of the expected-norm classification; the fixed-positive- p and growing-order assertions in this paper retain their own stated quantifiers and proofs.

A *core shape* means a shape every original connected component of which meets $U \cup V$. Such a shape has no isolated interior vertices, and every component floating after a separator has an attachment to that separator. For core shapes,

$$g = \frac{1}{2} \max_{S \in \mathcal{S}_{\min}} \#\{C \text{ component of } H - S : C \cap (U \cup V) = \emptyset\}.$$

The attachment qualification in (1.2) is essential for the full disconnected model. Most of the proof is carried out for core shapes; Section 9 reduces the general case to that one.

Theorem 1.4 (Strong upper moments). *Let t be the number of original connected components of H that miss both boundaries and contain at least one edge. For every fixed $A > 0$,*

$$\| \|M_{H,n}\| \|_{L^p} \leq C_{H,A} T_{H,n} p^{t/2}, \quad 1 \leq p \leq A \log(2n). \quad (1.5)$$

The same estimate holds for unbiased k_n -wise independent primitive signs whenever $k_n \geq K_{H,A} \log(2n)$, for a sufficiently large fixed constant $K_{H,A}$.

Theorem 1.4 controls moments of logarithmic order, not just fixed order, and the factor $p^{t/2}$ is not an artifact: an original boundary-free component with an edge is a scalar sum whose L^p size grows like $\sqrt{p}$, and t such factors are present. For $t = 0$, Markov’s inequality at logarithmic order gives arbitrarily small polynomial upper-tail probabilities at a sufficiently large constant multiple of $T_{H,n}$. For $t > 0$, the corresponding bound from (1.5) is at $T_{H,n} (\log(2n))^{t/2}$. The matching logarithmic lower bound in Theorem 1.3 is proved under full independence. No general transfer of that lower bound to limited independence is asserted.

1.4 Relation to prior work

The separator polynomial scale is established prior work. Ahn–Medarametla–Potechin give separator-based bounds up to polylogarithmic factors, including the isolated-interior correction [3, Theorems 2.24 and 6.1]; their Section 5 develops vertex partitioning for the trace argument. Bandeira–Lucca–Nizić–Nikolac–van Handel give a systematic coefficient-flattening treatment and derive graph-matrix bounds in Theorem 4.8 [1]. The earlier polynomial scale should not be attributed solely to this later treatment, and the present comparisons do not assert that every use of its stronger inequalities must lose a logarithm.

The exact polynomial-times-logarithm question is recorded as Randomstrasse Problem 31 [2]. Xu–Fu–Qiao–Tao [4, Theorem 1.3] give another proof of the expected-norm classification in precisely the dense-sign model of Theorem 1.3. Their Definition 1.1 fixes the global injectivity and boundary conventions. Their Appendix F.6 separately discusses growing moments; that discussion is not identified with an unrestricted all-order assertion here. No priority or independent-discovery claim is made.

The proof presented here uses the finite occupation representation, connected-creation commutators, critical-separator factorization, and conditional lower witnesses. This proof mechanism is distinct from the statement of the shared endpoint. No cited graph-matrix norm theorem is used as a black box below. The description “self-contained” refers to the supplied proofs, not to originality of the classical inequalities, Walsh representation, or every reduction.

1.5 Overview of the proof

The proof has an upper half and a lower half, which share a common first move (coloring) and a common combinatorial object (a minimum separator and the pieces of H that it cuts off). We describe the mechanism of each step here; the sections that follow carry it out.

Coloring (Section 3). Give every role x its own pool $P_x \subseteq [n]$ of labels, and keep only embeddings that send each role into its pool. This has three effects. Injectivity becomes automatic. A primitive sign W_{ij} now identifies the edge role xy it can represent, namely the one with $i \in P_x, j \in P_y$, so distinct edge roles use disjoint blocks of signs and the sign support of a monomial determines its embedding. And different pieces of the shape use independent signs, so products of pieces are products of independent random objects. The price is that we now have to return to the uncolored matrix. For the upper bound this is an average: a uniformly random coloring recovers $M_{H,n}$ exactly up to the constant h^h (Lemma 3.2). For the lower bound it is a contraction: conditional expectation and a signed Fourier average project the uncolored matrix onto a fixed coloring, and both operations can only decrease L^p norms (Lemma 3.3).

The upper bound: occupation operators (Sections 5–7). We bound $\mathbb{E} \operatorname{Tr}((MM^*)^q)$ for q of order $\log n$ and take $2q$ -th roots. A product of $2q$ monomials has nonzero expectation only if every sign occurs an even number of times. Instead of counting these cancellation patterns directly, we keep track of the parity state: the set Ω of primitive edges that currently occur an odd number of times. Multiplying by a monomial toggles its e edges in and out of Ω . In this picture M becomes a deterministic operator $\mathcal{W}$ on the space spanned by pairs (boundary assignment, Ω), and the expected trace is a sum of diagonal matrix elements of $\mathcal{W}^{2q}$ at $\Omega = \emptyset$. A closed walk of $2q$ steps that toggles e edges per step never reaches $|\Omega| > eq$, so $\mathcal{W}$ may be replaced by its compression to *grade* $|\Omega| \leq K := eq$ (Lemma 5.2). The problem is now to bound the norm of a deterministic operator on states of grade at most K , uniformly in the state.

Decompose $\mathcal{W}$ according to which of the shape’s edges land in Ω : the *occupied pattern* $F \subseteq E$. For a fixed pattern, the roles fall into three groups, see (5.3): roles all of whose edges are annihilated are *released*, roles none of whose edges are occupied are *fresh*, and the rest form S_F . Fresh roles carry free labels in the forward direction, released roles in the reverse direction, and every other label is pinned by the boundary or by the at most $2K$ labels present in Ω . Schur’s test bounds the norm of the pattern by $n^{(|R_F|+|Z_F|)/2}$ times a polynomial in K . The combinatorial key is that S_F is a U -to- V separator (Lemma 5.4), so $|R_F| + |Z_F| \leq h - s$: no pattern can beat $n^{(h-s)/2}$, and patterns with strict inequality are smaller by $n^{1/2}$ and only affect constants.

The *critical* patterns, those for which S_F is a minimum separator, are the whole story. For them the operator factors exactly as a product over the pieces of $H - S_F$ (Lemma 7.2): a side factor for the components meeting U , a side factor for those meeting V , and one factor per

floating component. The side factors are creation operators from a minimum separator and have norm $n^{\ell/2}$ up to a lower-order term (Lemma 6.4). Each floating factor is the creation, or annihilation, operator of a connected rooted scalar shape, and the central estimate of the upper bound is that such an operator, restricted to grade k , has norm about $\sqrt{k/e_J + 1} n^{c/2}$ (Lemma 6.2). The reason is an approximate canonical commutation relation: creating a copy of the shape and then removing one differs from removing one and then creating one only when the two copies overlap. Disjoint copies cancel exactly; identical copies give a diagonal term of size at most n^c ; and a *partially* overlapping pair is where connectedness enters, since a partially changed copy has an unchanged interior vertex adjacent to a changed one, whose label must lie in the support of Ω , and this costs a factor n against a polynomial in k . Thus $\mathcal{C}^*\mathcal{C} - \mathcal{C}\mathcal{C}^* \preceq n^c(1 + o(1))I$ on grade k , and, exactly as the boson relation $aa^* - a^*a = I$ gives $\|a^*\|_{\text{grade } k} = \sqrt{k+1}$, iterating along the grades gives $\|\mathcal{C}|_{\text{grade } k}\|^2 \leq (k/e_J + 1)n^c(1 + o(1))$.

Assembling, a critical pattern with separator S has norm at most $n^{(h-s)/2}(K+1)^{b_{\text{att}}(S)/2}$ up to constants and lower-order terms, the finitely many patterns sum to $n^{(h-s)/2}(q+1)^{b_*/2}$, and $2q$ -th roots at $q \asymp \log n$ give $(\log n)^{b_*/2}$ (Proposition 7.6). Note that the argument proves more than an expectation bound: it controls L^p norms for p up to a constant times $\log n$, and it uses only the expectation of a polynomial of degree $O(\log n)$ in the signs, which is what the limited-independence statement of Theorem 1.4 exploits.

The lower bound: a diagonal through the separator (Sections 4 and 8). Fix a minimum separator S with $b_{\text{att}}(S) = b_*$. For fixed separator labels A , the colored matrix factors as $M = LDR$ (Lemma 8.1): L is the matrix of the left side (rows U , columns S), R that of the right side, and D is diagonal in A with entry the product of the b_* floating fields $F_{j,A}$, times the separator-internal signs. The three factors are independent. If L and R were identities we would have $\|M\| = \max_A |D_{A,A}|$, a maximum over separator labels. In general we test M against the normalized columns $u_A = Le_A/\|Le_A\|$ and $v_A = R^*e_A/\|R^*e_A\|$ at the maximizing A . The cross terms $\langle Le_B, u_A \rangle$, $B \neq A$, vanish in expectation by a sign symmetry that exists precisely because S is minimal (Lemma 8.2), so

$$\mathbb{E}\|M\| \geq \alpha_L \alpha_R \mathbb{E} \max_A |D_{A,A}|,$$

where $\alpha_L \asymp m^{\ell/2}$ and $\alpha_R \asymp m^{r/2}$ are the expected column norms (Lemma 8.3). Since $\ell + r + \sum_j c_j = h - s$, it remains to show $\mathbb{E} \max_A |D_{A,A}| \gtrsim m^{(\sum_j c_j)/2} (\log m)^{b_*/2}$.

This is the classical extreme-value mechanism. Among m independent trials, an event of probability e^{-Cq} with $q = \delta \log m$ and $C\delta < 1$ occurs with probability tending to one. The event is that all b_* fields simultaneously exceed $\sqrt{q}$ times their standard size $m^{c_j/2}$, and its probability is controlled by Paley–Zygmund applied to F^{2q} , which requires the $2q$ -th and $4q$ -th moments of the fields (Proposition 4.2). The fields at different separator labels share their interior signs, so the trials are independent only conditionally on those signs; the conditional moments are random, and Proposition 4.4 shows that they concentrate around their means. Lemma 4.5 puts these together.

Completion (Sections 9–10). Original boundary-free components factor out as independent scalars, with L^p size $n^{c/2}\sqrt{p}$ and expected absolute value of order $n^{c/2}$: they add to the polynomial exponent and to the $p^{t/2}$ in Theorem 1.4, but not to the logarithmic exponent. Isolated interior vertices contribute the exact factor $(n - |V(H')|)_{i_0}$. Paley–Zygmund converts the first- and second-moment estimates into a lower bound for every fixed positive L^p size. Finally, since the upper argument only uses expectations of polynomials of logarithmic degree, it transfers to logarithmically-wise independent signs.

Map of the proof. The table records what each section establishes and which half of the theorem uses it.

Section	Content	Used by
2	Schur test; Paley–Zygmund; fourth moments of sign polynomials	both halves
3	Harmless reductions; the colored model; color average (Lemma 3.2); Fourier projection (Lemma 3.3)	both halves
4	Moments of a connected scalar field; the two-replica conditional count; simultaneous excursions	lower half; Section 9
5	Walsh occupation representation; vacuum compression; occupied patterns and their signatures	upper half
6	Connected creation commutator; side creation from a minimum separator	upper half
7	Exact factorization of critical patterns; core upper bound at logarithmic moments	upper half
8	LDR factorization; normalized-column symmetry; column scale; core lower bound	lower half
9	Boundary-free components; isolated interiors; all fixed $p > 0$	both halves
10	Upper tails; limited independence	consequences
11	Examples; scope	—

Two examples in miniature. The shape (b) of the introduction, the edge rx with $U = V = (r)$, is small enough that every step of both halves can be written out in a few lines. We return to it in Example 6.1 (the creation commutator is exact), Example 7.3 (the occupation operator is $\mathcal{C} + \mathcal{C}^*$ and the upper bound is $(n \log n)^{1/2}$ in two lines), and Example 8.5 (the lower bound is the extreme value of m independent sums). The shape (a), the edge uv with $U = (u)$, $V = (v)$, is treated in Example 7.5, where the occupation machinery reproduces the order $n^{1/2}$ of a Wigner matrix.

1.6 Notation and conventions

We write $[n] = \{1, \dots, n\}$ and $(m)_q = m(m-1) \cdots (m-q+1)$, with $(m)_q = 0$ if $q > m$. The double factorial $d_q = (2q-1)!!$, with $d_0 = 1$, is the number of perfect matchings of $2q$ points. The norm $\|\cdot\|$ is the Euclidean norm of a vector or the operator norm of a matrix; $\langle \cdot, \cdot \rangle$ is the Euclidean inner product; $\preceq$ is the Loewner order on self-adjoint matrices. For $p > 0$ and a real random variable X , $\|X\|_{L^p} = (\mathbb{E}|X|^p)^{1/p}$. All random spaces used in the main proof are finite sign spaces, so all expectations are finite sums.

A *primitive sign*, or *primitive edge*, is one of the variables W_{ij} , identified with the unordered pair $\{i, j\}$ of ground labels. The *edge roles* are the edges $xy \in E$ of the shape; an embedding ϕ sends the edge role xy to the primitive edge $\{\phi(x), \phi(y)\}$. Constants c_H, C_H depend only on the shape, and $C_{H,A}$ on the shape and the parameter A ; they may change from line to line. We write $a \asymp b$ if $c \leq a/b \leq C$ for such constants.

Throughout, “for all sufficiently large n ” and “eventually” mean for $n \geq n_0(H)$, or $n \geq n_0(H, A)$ when a parameter A is present. When both m and n appear, m is a pool size and $m \asymp_H n$.

2 Elementary inequalities used in the proof

This section collects three classical inequalities, with proofs of the forms used here. The Schur test [8] is the only tool used to bound operator norms of the deterministic occupation operators in Sections 5–7; Paley–Zygmund [7] is the only tool used to convert moment estimates into probability lower bounds; and the fourth-moment estimate is what allows us to pass from L^2 to L^1 sizes of fixed-degree sign polynomials, which is needed for the column norms of Section 8 and the scalar factors of Section 9.

We write $\|X\|_{L^p} = (\mathbb{E}|X|^p)^{1/p}$ for $p > 0$. For $p \geq 1$ this is a norm. We use its triangle inequality, monotonicity in p on probability spaces, and Hölder’s inequality. All random spaces used in the main proof are finite sign spaces.

Lemma 2.1 (Schur bound). *If a finite real matrix $A = (a_{ij})$ has absolute row sums at most r and absolute column sums at most c , then $\|A\| \leq \sqrt{rc}$.*

Proof. For every vector x , weighted Cauchy–Schwarz gives

$$\sum_i \left| \sum_j a_{ij} x_j \right|^2 \leq \sum_i \left(\sum_j |a_{ij}| \right) \left(\sum_j |a_{ij}| |x_j|^2 \right) \leq r \sum_j |x_j|^2 \sum_i |a_{ij}| \leq rc \|x\|^2. \quad \square$$

The Schur bound is crude for dense matrices but exact for permutation-like matrices, and the occupation operators below are sums of partial permutations with nonnegative integer coefficients. Every application of it will be of the following form: for a fixed source state, count the number of target states with multiplicity, and vice versa.

Lemma 2.2 (Paley–Zygmund). *For $Y \geq 0$ with finite second moment and $0 < \theta < 1$,*

$$\mathbb{P}\{Y \geq \theta \mathbb{E}Y\} \geq (1 - \theta)^2 \frac{(\mathbb{E}Y)^2}{\mathbb{E}Y^2}.$$

Proof. The contribution of $\{Y < \theta \mathbb{E}Y\}$ to $\mathbb{E}Y$ is at most $\theta \mathbb{E}Y$, so the contribution of the complementary event is at least $(1 - \theta)\mathbb{E}Y$. Cauchy–Schwarz bounds that contribution by $(\mathbb{E}Y^2)^{1/2} \mathbb{P}\{Y \geq \theta \mathbb{E}Y\}^{1/2}$. Squaring gives the claim. $\square$

Lemma 2.3 (Fourth moments of fixed-degree sign polynomials). *If P is a real polynomial of independent signs, reduced to a multilinear polynomial of degree at most d , then*

$$\|P\|_{L^4} \leq 3^{d/2} \|P\|_{L^2}.$$

If X is a finite real vector whose coordinates have degree at most d , then

$$(\mathbb{E}\|X\|_2^4)^{1/2} \leq 3^d \mathbb{E}\|X\|_2^2, \quad \mathbb{E}\|X\|_2 \geq c_d (\mathbb{E}\|X\|_2^2)^{1/2}, \quad (2.1)$$

where one may take $c_d = (4\sqrt{2}3^{2d})^{-1}$.

The first assertion is the classical Bonami–Beckner hypercontractive inequality at exponents 2 and 4 [5, 6]; we include the short inductive proof because the explicit constant is used. The second assertion says that a vector of bounded-degree sign polynomials has expected norm comparable to its root-mean-square norm.

Proof. Write the Walsh expansion $P = \sum_J \hat{P}(J) \chi_J$, where $\chi_J = \prod_{j \in J} \varepsilon_j$ runs over subsets J of the signs, and put

$$Q(P) = \sum_J 3^{|J|} \hat{P}(J)^2.$$

We prove $\mathbb{E}P^4 \leq Q(P)^2$ by induction on the number of signs. If P involves no sign it is a constant and both sides equal P^4 . Otherwise single out one sign ε and write $P = g + \varepsilon h$, where

g and h do not involve ε . Since ε is symmetric and independent of (g, h) , the odd powers of ε average to zero and

$$\mathbb{E}P^4 = \mathbb{E}g^4 + 6\mathbb{E}g^2h^2 + \mathbb{E}h^4.$$

The induction hypothesis gives $\mathbb{E}g^4 \leq Q(g)^2$ and $\mathbb{E}h^4 \leq Q(h)^2$, and Cauchy–Schwarz gives $\mathbb{E}g^2h^2 \leq (\mathbb{E}g^4)^{1/2}(\mathbb{E}h^4)^{1/2} \leq Q(g)Q(h)$. Hence

$$\mathbb{E}P^4 \leq Q(g)^2 + 6Q(g)Q(h) + Q(h)^2 \leq (Q(g) + 3Q(h))^2.$$

The last expression equals $Q(P)^2$: the Walsh coefficients of P at sets not containing ε are those of g , and those at sets $J \ni \varepsilon$ are the coefficients of h at $J \setminus \{\varepsilon\}$, with the weight $3^{|J|}$ larger by the factor 3. This proves $\mathbb{E}P^4 \leq Q(P)^2$. Orthogonality of distinct Walsh characters gives $\mathbb{E}P^2 = \sum_J \hat{P}(J)^2$, and since all sets J with nonzero coefficient have $|J| \leq d$, $Q(P) \leq 3^d \mathbb{E}P^2$. Taking fourth and square roots gives the first assertion.

For vectors, the triangle inequality in L^2 and the first assertion applied to each coordinate give

$$(\mathbb{E}(\sum_i X_i^2)^2)^{1/2} \leq \sum_i (\mathbb{E}X_i^4)^{1/2} \leq 3^d \sum_i \mathbb{E}X_i^2 = 3^d \mathbb{E}\|X\|_2^2.$$

Now apply Lemma 2.2 to $Y = \|X\|_2^2$ with $\theta = 1/2$: the probability that $\|X\|_2^2 \geq \frac{1}{2}\mathbb{E}\|X\|_2^2$ is at least $\frac{1}{4}(\mathbb{E}Y)^2/\mathbb{E}Y^2 \geq \frac{1}{4}3^{-2d}$. On that event $\|X\|_2 \geq (\frac{1}{2}\mathbb{E}\|X\|_2^2)^{1/2}$, which gives the second assertion with the stated constant. $\square$

3 Graph reductions and exact role coloring

This section performs two preparatory steps. First, three harmless simplifications of the shape (boundary-internal edges, original boundary-free components, isolated interior vertices) are shown to change the matrix in controlled ways and the exponents not at all. Second, the colored model is introduced, and the two lemmas that connect it to the original model are proved: an exact averaging identity, which transfers upper bounds, and a contractive Fourier projection, which transfers lower bounds.

Vertex partitioning has an established role in graph-matrix moment bounds [3, Section 5]. Related typed/global model transfers are also given in [4, Section 2]. The exact averaging and Fourier projection needed for this paper’s conventions are proved here; the coloring device itself is not asserted to be new.

3.1 Elementary separator facts

Lemma 3.1 (Reductions). *The following operations have the stated effects.*

- (i) *Deleting all edges internal to U or internal to V preserves the family of separators, the minimum-separator size, and every attached floating-component count at a minimum separator. It changes the graph matrix only by multiplication on the left and right by diagonal sign matrices.*
- (ii) *No minimum separator uses a vertex of an original component missing both boundaries. Deleting all such components preserves the minimum-separator family and all attached floating-component counts.*
- (iii) *If H' is obtained by deleting the i_0 isolated interior vertices, then, for $n \geq h$,*

$$M_{H,n} = (n - |V(H')|)_{i_0} M_{H',n}. \quad (3.1)$$

The separator size and attached floating counts are unchanged.

Item (i) is what lets us assume, in the operator arguments of Sections 7 and 8, that the boundary sets are independent sets of the graph; the deleted edges are functions of the row assignment alone or the column assignment alone, and diagonal sign matrices do not change operator norms. Items (ii) and (iii) justify treating the general shape by first treating its core.

Proof. For (i), if a U -to- V path avoids a proposed separator, truncate it at its first visit to V , and then at its last preceding visit to U . The resulting path has no boundary vertex in its interior. It therefore uses no edge internal to either boundary. A path of length zero needs no change. Hence a set separates U from V in the original graph if and only if it does so after the deletion; this proves separator preservation.

A component of the edge-deleted graph missing both boundaries has no endpoint of a deleted edge. It was consequently already a component before the edge deletion, and the converse holds. The edges from such a component to a separator are also unaffected. This proves the count assertion. Each removed edge internal to U is a sign depending only on the row assignment, and each remaining removed edge internal to V depends only on the column assignment. Assign an edge internal to both boundaries to just one factor. These diagonal factors have absolute entries one, proving norm preservation. No interior vertex becomes isolated by this operation.

For (ii), a vertex in an original boundary-free component is irrelevant to every boundary-to-boundary path. Its removal from a separator would leave a separator of smaller cardinality. Thus a minimum separator avoids all those components. Deleting them leaves exactly the same minimum separators. They have no edge to such a separator, so they are never counted in (1.2).

For (iii), every embedding of H' extends to the deleted distinguished roles in exactly $(n - |V(H')|)_{i_0}$ ordered ways, since the i_0 deleted roles must receive distinct labels outside the image of H' . No monomial changes, because an isolated role has no edges. The separator assertion follows also from (ii). $\square$

3.2 The colored model

Fix pairwise disjoint pools $P_x \subseteq [n]$, one for each role $x \in \mathcal{V}$. The colored matrix M_H^{col} is obtained by retaining only those terms in (1.1) with $\phi(x) \in P_x$. Coloring enforces global injectivity automatically.

Different abstract edge roles use disjoint blocks of primitive signs:

$$\mathcal{B}_{xy} = \{\{i, j\} : i \in P_x, j \in P_y\}, \quad xy \in E.$$

In particular, primitive edges belonging to different edge roles cannot coincide: a primitive edge $\{i, j\}$ determines the pair of pools containing i and j , and hence the edge role. An edge support determines every nonisolated role label. An isolated role label is also determined if it belongs to a fixed boundary assignment.

These are the properties of coloring that the rest of the proof uses. The first (disjoint sign blocks) is what makes the cancellation structure of moments transparent; the second (supports determine embeddings) is used in every Gram-matrix computation; and the independence of signs across disjoint sets of edge roles is what makes the factorizations of Sections 8 and 9 products of independent factors.

Lemma 3.2 (Upper color average). *For $h > 0$, assign each ground label independently a uniform color from the h roles, and take P_x to be its color class. Then*

$$M_{H,n} = h^h \mathbb{E}_{\text{colors}} M_H^{\text{col}}. \quad (3.2)$$

Consequently a uniform colored upper bound on $\|M_H^{\text{col}}\|_{L^p}$, $p \geq 1$, transfers to the uncolored matrix, at the cost of the fixed factor h^h .

Proof. Fix an injective embedding ϕ . It survives the coloring exactly when each of its h labels $\phi(x)$ receives the color x , which happens with probability h^{-h} since the colors of distinct labels are independent. Taking the average of (1.1) term by term gives (3.2). The bound follows from the triangle inequality for the L^p norm of a matrix norm, since the average over colors is a convex combination. All colored matrices are viewed as zero-padded matrices on the original boundary spaces. $\square$

Note that (3.2) is an identity between polynomials in the signs, valid for every realization; this is why it transfers to limited independence in Section 10. The pools produced by random coloring have unequal sizes, which is why the upper bounds below are proved for arbitrary pools of sizes at most n .

Lemma 3.3 (Fourier projection to one coloring). *Suppose H has no isolated interior vertices, and fix arbitrary pairwise disjoint pools. Let*

$$a_H = |\text{Aut}(H; U \cup V)|,$$

where the automorphisms fix each boundary role pointwise. For $p \geq 1$,

$$\| \| M_{H,n} \| \|_{L^p} \geq a_H \| \| M_H^{\text{col}} \| \|_{L^p}. \quad (3.3)$$

The lemma says that the uncolored matrix is at least as large, in every L^p sense, as any fixed coloring of it. The proof exhibits the colored matrix as the image of the uncolored one under three norm-nonincreasing operations: compression of the boundary spaces to the boundary pools, averaging over the signs that no colored embedding uses, and a signed average over block-wise sign flips that picks out the monomials using each edge block exactly once. The only surprise is that the surviving monomials are not just the colored embeddings but also their images under boundary-fixing automorphisms, which is where the factor a_H comes from.

Proof. Compress the row and column spaces to their prescribed boundary pools, giving a random matrix N . Average over every primitive sign outside the allowed blocks $\mathcal{B}_{xy}$, and call the result N_0 . Each graph monomial uses distinct signs, so it survives this average exactly when all its primitive edges lie in allowed blocks.

For a vector $\sigma \in \{\pm 1\}^E$, let W^σ multiply every primitive sign in block $\mathcal{B}_{xy}$ by σ_{xy} . Apply the signed average

$$\mathcal{Q}N_0(W) = 2^{-e} \sum_{\sigma \in \{\pm 1\}^E} \left(\prod_{xy \in E} \sigma_{xy} \right) N_0(W^\sigma). \quad (3.4)$$

A monomial with k_{xy} primitive edges in block $\mathcal{B}_{xy}$ is multiplied by $\prod_{xy} \sigma_{xy}^{k_{xy}+1}$, whose average over σ is one if every k_{xy} is odd and zero otherwise. Thus $\mathcal{Q}$ selects precisely the monomials with odd degree in each allowed block. Their total degree is e , so they use each of the e blocks exactly once.

Consider such a surviving monomial, coming from an uncolored embedding ϕ . Every nonisolated target pool is an endpoint of a used block and is therefore occupied by some $\phi(x)$. The pools of isolated boundary roles are occupied by boundary compression. There are no other isolated roles. Thus the h source roles occupy all h pools, and their role-to-pool map $x \mapsto (\text{pool containing } \phi(x))$ is a bijection. It sends the edge set onto itself, because each edge role of ϕ lands in exactly one block, and it fixes every boundary role. It is a boundary-fixing automorphism.

Conversely, a colored embedding reindexed by any such automorphism survives the two averages. Each contributes the same colored monomial with the same boundary indices. Reindexing the interior summations therefore gives the exact matrix identity

$$\mathcal{Q}N_0 = a_H M_H^{\text{col}}. \quad (3.5)$$

This argument does not require equal pool sizes. For a fixed automorphism, the reindexing is a bijection between its surviving embeddings and the prescribed colored embeddings.

Compression contracts the operator norm. Conditional expectation contracts its L^p size by Jensen. Finally, each block flip preserves the full independent sign law, so the triangle inequality applied to (3.4) gives $\|\mathcal{Q}N_0\|_{L^p} \leq \|N_0\|_{L^p}$. Together with (3.5), this proves the lemma. If $e = 0$, all roles are boundary roles and the same argument has empty sign averages. $\square$

Example 3.4 (The automorphism factor). Let H consist of two paths $a x_1 b$ and $a x_2 b$ sharing the endpoints, with $U = (a)$, $V = (b)$, and pools P_a, P_b, P_1, P_2 . An uncolored monomial $W_{ai}W_{ib}W_{aj}W_{jb}$ with $a \in P_a$, $b \in P_b$ survives the two averages exactly when one of i, j lies in P_1 and the other in P_2 : with $i \in P_1$, $j \in P_2$ it is the colored embedding itself, and with $i \in P_2$, $j \in P_1$ it is the colored embedding composed with the automorphism exchanging x_1 and x_2 . Both terms have the same boundary indices and the same value, so $\mathcal{Q}N_0 = 2M_H^{\text{col}}$, and indeed $a_H = 2$. For the single path $a x b$ there is no nontrivial boundary-fixing automorphism and $\mathcal{Q}N_0 = M_H^{\text{col}}$.

Remark 3.5. Equation (3.3) is a moment comparison. It is not a pointwise norm comparison for a fixed realization of W . Full independence is used in the conditional expectation and sign-law invariance steps. This is the reason the lower bound of Theorem 1.3 is stated under full independence only.

4 Connected scalar fields and conditional excursions

This section studies the scalar random variables that the floating components of a separator produce: for a connected graph attached to a set of root roles, the sum over colored placements of its interior of the product of its edge signs. Three results are proved. Proposition 4.2 computes the even moments of such a field up to a $1 + o(1)$ factor for moment orders up to any fixed power of $\log m$. Proposition 4.4 shows that the *conditional* moments, given the signs internal to the interior, concentrate around the unconditional ones. Lemma 4.5 then shows that, among the m root labels of a pool, some label makes all the fields attached to it simultaneously large, with an excess of $(\log m)^{1/2}$ per field. The last result is what produces the logarithmic factor in the lower bound; the first result is also used, through Corollary 4.3, for the boundary-free components in Section 9.

4.1 A finite colored moment count

Let J be a fixed simple graph with root roles R and nonempty interior C , where $c = |C|$. Assume $J[C]$ is connected and every interior role has positive degree. Remove all root-internal edges. All roles have disjoint pools. Fix a root-label assignment A , and define the colored scalar field

$$F_A = \sum_{\phi(x) \in P_x, x \in C} \prod_{xy \in E(J)} W_{\phi(x)\phi(y)}, \quad \phi|_R = A. \quad (4.1)$$

Unused roots cause no difficulty. A rootless connected graph is allowed if it has an edge. An isolated interior singleton is excluded.

Example 4.1 (Two small fields). The simplest field has one root r , one interior role x , and the edge rx : with $A = a$ and $P_x = P$,

$$F_a = \sum_{j \in P} W_{aj},$$

a sum of $m = |P|$ independent signs. Its even moments are $\mathbb{E}F_a^{2q} = \sum_d N_d(m)_{q-d}$, where N_d counts the set partitions of $[2q]$ into $q-d$ blocks of even size: a sequence $(j_1, \dots, j_{2q})$ contributes if and only if every label occurs an even number of times, and the partition records which

positions carry equal labels. The leading term $d = 0$ comes from partitions into q pairs, of which there are d_q , each with $(m)_q$ labelings.

The next simplest field has one root r , interior path xy , and edges rx and xy :

$$F_a = \sum_{i \in P_x} \sum_{j \in P_y} W_{ai} W_{ij} = \sum_{i \in P_x} W_{ai} \sigma_i, \quad \sigma_i = \sum_{j \in P_y} W_{ij}.$$

Here the fields F_a for different root labels a share the interior signs W_{ij} ; only the root-to-interior signs W_{ai} are private to the label a . Conditionally on the interior signs, F_a is a weighted sum of fresh independent signs with the random weights σ_i , and its conditional moments depend on those weights: for instance its conditional variance is $\sum_i \sigma_i^2$, a random quantity with mean $|P_x||P_y|$. This dependence on the interior signs is the reason for the two-replica count in Section 4.2.

Put

$$d_q = (2q - 1)!!, \quad d_0 = 1, \quad B_c = 6c + 8.$$

All coefficients in the moment expansion are nonnegative: a product of independent signs has expectation zero or one.

Proposition 4.2 (Finite scalar moments). *If all interior pools have size at most n , and $\rho = (2cq)^{B_c}/n < 1$, then*

$$\mathbb{E} F_A^{2q} \leq \frac{d_q}{1 - \rho} n^{cq}. \quad (4.2)$$

If all interior pools have the same size m , then also

$$\mathbb{E} F_A^{2q} \geq d_q ((m)_q)^c. \quad (4.3)$$

Here $(m)_q = m(m-1) \cdots (m-q+1)$, interpreted as zero if $q > m$. Consequently, for every fixed $B > 0$, uniformly for $1 \leq q \leq (\log m)^B$,

$$\mathbb{E} F_A^{2q} = (1 + o(1)) d_q m^{cq} \quad (4.4)$$

on equal pools.

The proposition says that, at any polylogarithmic moment order, the field behaves like a Gaussian of variance m^c : $d_q m^{cq}$ is the $2q$ -th moment of such a Gaussian. The proof is a count of the *equality patterns* of $2q$ embeddings, that is, of which interior labels coincide across the $2q$ copies. The dominant patterns are those in which the copies come in identical pairs with distinct labels across pairs, and there are d_q ways to pair them. Every other pattern has fewer distinct labels, hence fewer realizations by a power of n , and the point of the proof is that the number of such patterns grows only polynomially in q per lost label, so that the loss of n wins as soon as n exceeds a fixed power of q . Connectedness of the interior is what forces the copies to pair up as wholes.

Proof. Step 1: parity, and a lower bound on multiplicities. Expand the moment over $2q$ ordered colored embeddings. In a surviving term every primitive sign occurs an even number of times. Every interior label must occur in at least two copies: otherwise an edge incident to its unique occurrence would be a singleton primitive edge, since role colors prevent a primitive edge from being represented by two different edge roles.

Step 2: the defect and the marked copies. Let v be the number of distinct interior labels across all copies, and define the defect $d = cq - v \geq 0$; the total number of interior label occurrences

is $2cq$, and $v \leq cq$ because every label occurs at least twice. If k_λ is the number of copies containing label λ , then

$$\sum_{\lambda} (k_\lambda - 2) = 2cq - 2v = 2d.$$

Mark each copy containing a label of multiplicity at least three. Since $k \leq 3(k-2)$ for $k \geq 3$, the number r of marked copies obeys

$$r \leq \sum_{k_\lambda \geq 3} k_\lambda \leq 6d. \quad (4.5)$$

Thus a pattern of small defect has few marked copies, and all other copies contain only labels of multiplicity exactly two.

Step 3: unmarked copies pair up as wholes. Consider an unmarked copy. An interior label x in it occurs in just one other copy. Every edge of the first copy incident to x must recur in that other copy, since the primitive edge must occur an even number of times and the only other occurrence of x is there. Propagating along the connected graph $J[C]$ shows that all interior labels of the first copy occur in that same other copy. This includes the singleton case, using a root attachment. The other copy has exactly c interior roles, so its complete interior image set is the same and it too is unmarked. Role colors identify the role of each label, so the two embeddings agree role by role. They form an isolated whole-copy pair: no third copy uses any of their labels. All unmarked copies are partitioned into such pairs, which we call *clean pairs*.

Step 4: the count at defect zero. Count equality patterns canonically by scanning copy positions and then the fixed role order, naming each new label at its first appearance. At defect zero there are no marked copies, so every copy belongs to a clean pair, and there are exactly d_q patterns, one per pairing of the $2q$ copy positions. For equal pools each has $((m)_q)^c$ labelings: each role receives q distinct labels from its pool, one per pair. This proves the lower bound (4.3), since all terms are nonnegative.

Step 5: the count at positive defect. For positive defect, encode the marked copy positions, a pairing of the unmarked positions, and an arbitrary partition of the cr marked role occurrences. Each first unmarked copy introduces new labels; its partner repeats them. The marked labels are disjoint from these clean-pair labels. Thus this is a reconstructive overcount of the canonical patterns. The number $N_{q,d}$ of patterns of defect $d \geq 1$ satisfies

$$N_{q,d} \leq d_q \sum_{0 \leq r \leq \min(2q, 6d)} \binom{2q}{r} (cr)^{cr},$$

with $0^0 = 1$. The pairing factor of any smaller even set is at most d_q ; impossible parity choices only enlarge the bound. Each summand is at most $(2cq)^{(c+1)r} \leq (2cq)^{6(c+1)d}$. There are at most $2q+1 \leq (2cq)^2$ summands. Hence

$$N_{q,d} \leq d_q (2cq)^{(6c+8)d}. \quad (4.6)$$

Allowing arbitrary marked partitions, including partitions incompatible with role colors, is only an overcount.

Step 6: summation. A pattern with $cq - d$ labels has at most n^{cq-d} realizations on arbitrary pools of sizes at most n . Summing (4.6) over $d \geq 0$ as a geometric series with ratio ρ gives (4.2). Finally,

$$\frac{(m)_q}{m^q} = \prod_{j=0}^{q-1} (1 - j/m) = 1 + O(q^2/m)$$

uniformly when $q^2/m \rightarrow 0$, and $(2cq)^{Bc}/m \rightarrow 0$ throughout any fixed polylogarithmic range. Combining the upper and lower bounds proves (4.4). $\square$

Corollary 4.3 (Scalar upper moments). *For arbitrary pools of sizes at most n , every fixed $A_0 > 0$, and all sufficiently large n ,*

$$\|F_A\|_{L^p} \leq C_{J,A_0} n^{c/2} \sqrt{p}, \quad 1 \leq p \leq A_0 \log(2n).$$

The bound is uniform in the fixed root labels. A rootless field obeys the same estimate without any root assignment.

Proof. Choose an integer $q \geq 1$ with $2q \geq p$ and $2q \leq p + 2$. Use $d_q \leq (2q)^q$ in (4.2) and monotonicity of moments: $\|F_A\|_{L^p} \leq \|F_A\|_{L^{2q}} \leq (d_q/(1-\rho))^{1/(2q)} n^{c/2} \leq C\sqrt{2q} n^{c/2}$. The geometric denominator is bounded away from zero throughout the specified range, since $q = O(\log n)$ there. $\square$

The $\sqrt{p}$ in the corollary is the growth rate of the L^p norm of a Gaussian, and it is sharp: the one-vertex field of Example 4.1 is a sum of independent signs.

4.2 The two-replica conditional count

Suppose now that J has at least one root-to-interior edge, and that all interior pools have size m . Separate the signs in (4.1) into interior signs ω and root-to-interior signs x . Put

$$Z_q(\omega) = \mathbb{E}_x F_A(x, \omega)^{2q}, \quad \mu_q = \mathbb{E}_\omega Z_q.$$

The conditional law, and hence $Z_q(\omega)$, is the same for every fixed legal root assignment: each root selects a fresh independent sign array of the same dimensions.

The random variable $Z_q(\omega)$ is the $2q$ -th moment of the field given the interior signs. In Example 4.1, for the path field, $Z_1(\omega) = \sum_i \sigma_i^2$. The lower-bound argument needs $Z_q(\omega)$ to be close to its mean for all q up to order $\log m$, simultaneously, with probability tending to one. The next proposition provides this through a second-moment bound on Z_q , whose proof is a moment count with $4q$ copies in two groups, the groups sharing the interior signs but using independent root-to-interior signs.

Proposition 4.4 (Conditional moment concentration). *For every fixed $B > 0$, uniformly for $1 \leq q \leq (\log m)^B$,*

$$\mu_q = (1 + o(1)) d_q m^{cq}, \quad \frac{\text{Var } Z_q}{\mu_q^2} \leq C_c \frac{(4cq)^{B_c}}{m} \quad (4.7)$$

for all sufficiently large m .

Proof. The first statement is Proposition 4.2, since $\mu_q = \mathbb{E} F_A^{2q}$. For the second, represent $\mathbb{E} Z_q^2$ by $4q$ copies in two groups of $2q$. The groups share the interior signs ω , but have independent replicas of *all* root-to-interior arrays. A surviving product must have even multiplicity for each interior primitive sign and for each root-edge sign together with its replica index.

The multiplicity and marking argument in the previous proof applies unchanged, now with defect $d = 2cq - v$ and at most $6d$ marked copies. An unmarked copy again pairs with a single other copy along its connected interior. Since there is an attachment edge, its repeated root-edge sign forces that partner to be in the same replica group. Thus every clean pair stays within its group.

At defect zero the exact contribution is

$$d_q^2 ((m)_{2q})^c. \quad (4.8)$$

Indeed the two group pairings can be chosen in d_q^2 ways, and the $2q$ pair labels for each role must be distinct across both groups. At positive defect choose marked positions among $4q$ positions, pair unmarked positions within their groups, and store a partition of the marked

occurrences. The group pairing factor is at most d_q^2 . The previous count, with $2q$ replaced by $4q$, gives

$$N_{q,d}^{(2)} \leq d_q^2 (4cq)^{B_c d}.$$

Therefore, whenever $\rho_2 = (4cq)^{B_c}/m < 1$,

$$\mathbb{E} Z_q^2 \leq \frac{d_q^2 m^{2cq}}{1 - \rho_2}.$$

On the other hand, $\mu_q \geq d_q((m)_q)^c$ by (4.3). For q in the specified range, falling-factorial estimates yield

$$\frac{\text{Var } Z_q}{\mu_q^2} = \frac{\mathbb{E} Z_q^2}{\mu_q^2} - 1 \leq \frac{1}{1 - \rho_2} \left(\frac{m^q}{(m)_q} \right)^{2c} - 1 \leq C_c \left(\rho_2 + \frac{q^2}{m} \right) \leq C_c \rho_2.$$

This proves the claim, without assuming any concentration of the individual interior coefficients. $\square$

The mechanism deserves a comment. The expansion of $\mathbb{E} Z_q^2$ contains the “product” patterns, in which the two groups pair up separately, and these contribute essentially μ_q^2 ; every other pattern has a label shared between the groups or a label of multiplicity at least three, hence a positive defect and a loss of at least one factor of m . What the attachment edge guarantees is that a clean pair cannot straddle the two groups: the root-edge sign of a copy carries its replica index, and a partner in the other group could not repeat it.

4.3 Simultaneous component excursions

Lemma 4.5 (Simultaneous component excursions). *Let $C_1, \dots, C_b$ be finitely many nonempty connected interiors, with respective sizes $c_1, \dots, c_b$, each attached by at least one edge to a common root-role set S . Give every role a disjoint pool of size m , and let $F_{j,A}$ be its field at root tuple A . For $b \geq 1$,*

$$\mathbb{P} \left\{ \max_A \prod_{j=1}^b |F_{j,A}| \geq c_H m^{(c_1 + \dots + c_b)/2} (\log m)^{b/2} \right\} \longrightarrow 1. \quad (4.9)$$

For $b = 0$, the product and its maximum are defined to be one.

The lemma is the source of the logarithmic factor. Its content is the familiar extreme-value fact that the maximum of m independent approximately Gaussian variables of standard deviation σ is of order $\sigma(\log m)^{1/2}$, applied simultaneously to b fields. The proof implements this at the level of moments: at moment order $q \asymp \log m$, Paley–Zygmund shows that a single field exceeds $\sqrt{q}$ times its standard size with probability at least $e^{-Cq} = m^{-C\delta}$, which is much larger than $1/m$ if δ is small; so among m independent trials one succeeds. Two features of the situation require care. First, the fields at different root tuples are independent only conditionally on the interior signs, so the argument is run conditionally, and Proposition 4.4 guarantees that the conditional moments are typical. Second, the b components must succeed at the *same* root tuple, which is arranged by using synchronous tuples and the conditional independence of the components.

Proof. Step 1: the interior signs are typical. Condition on the interior signs of every component. Apply Proposition 4.4 to each component at $q = \lfloor \delta \log m \rfloor$ and $2q$, where $\delta > 0$ will be chosen below. Chebyshev’s inequality, the variance estimate, and a finite union bound show that with probability tending to one over the interiors, simultaneously for every component,

$$Z_q \geq \tfrac{1}{2} d_q m^{cq}, \quad Z_{2q} \leq 2 d_{2q} m^{2cq}.$$

Step 2: one trial succeeds with probability $e^{-O(q)}$. On this event, conditional Paley–Zygmund for F^{2q} , whose conditional mean is Z_q and conditional second moment is Z_{2q} , gives with $\theta = 1/2$

$$\mathbb{P}_x\{|F| \geq (Z_q/2)^{1/(2q)} \mid \omega\} \geq \frac{Z_q^2}{4Z_{2q}} \geq \frac{d_q^2}{32d_{2q}}. \quad (4.10)$$

Elementary factorial bounds suffice here. Since $d_q \geq q! \geq (q/e)^q$ and $d_{2q} \leq (4q)^{2q}$, there is an absolute C_0 such that the last expression is at least $\exp(-C_0q)$ for $q \geq 1$. Also $(Z_q/2)^{1/(2q)} \geq c_0 m^{c/2} \sqrt{q}$ for an absolute $c_0 > 0$, since $(d_q/4)^{1/(2q)} \geq \frac{1}{2}(q/e)^{1/2}$.

Step 3: m independent synchronous trials. Enumerate the labels of each root pool as $\lambda_z(1), \dots, \lambda_z(m)$. Use the m synchronous tuples

$$A^{(i)} = (\lambda_z(i))_{z \in S}, \quad 1 \leq i \leq m.$$

Conditional on the interiors, the root-to-interior arrays are disjoint for different components, and disjoint for different tuples $A^{(i)}$. Thus all these trial arrays are independent. At one tuple, the probability of the required excursions in all b components is at least $\exp(-C_0 b q)$, by conditional independence across components.

Choose δ so that $C_0 b \delta < 1/2$. Eventually the failure probability across all m independent conditional tuples is at most

$$(1 - \exp(-C_0 b q))^m \leq \exp(-m e^{-C_0 b q}) \leq \exp(-m^{1/2}).$$

On success, $\prod_j |F_{j,A}| \geq \prod_j c_0 m^{c_j/2} \sqrt{q}$ at the successful tuple. Include the probability of the good-interior event and use $q \asymp \log m$ to prove (4.9). $\square$

Remark 4.6. The constant c_H in (4.9) depends on δ , hence on b : the more components must succeed simultaneously, the rarer the joint event and the smaller the admissible δ . This is harmless because only the order $(\log m)^{b/2}$ matters, but it means that the argument does not identify the constant in front of the logarithm, and it should not be expected to: the true extreme value of correlated fields is a finer question.

5 The exact Walsh occupation representation

This section sets up the operator model for the upper bound. The idea is to replace the random matrix by a deterministic operator that keeps track of which primitive signs have been used an odd number of times. Expectation then becomes a single matrix element, moments become powers, and the problem becomes a norm bound for a deterministic operator on a finite-dimensional space. The construction is elementary linear algebra on a finite probability space; the words “occupation,” “grade,” and “vacuum” name basis vectors and an integer attached to them, and carry no further meaning. Walsh–Fourier analysis on product sign spaces is classical [9]; more general vacuum representations of moments occur in the interacting-Fock-space literature [10]. The contribution of the subsequent sections is in estimating the structured operators, not in the multiplication representation alone.

5.1 Walsh characters as occupation states

Throughout the operator argument all roles have disjoint pools of sizes at most n . Let $\mathcal{E}$ be the set of all unordered pairs of ground labels, including pairs unused by the shape. On the independent sign space the Walsh characters $\chi_\Omega = \prod_{a \in \Omega} W_a$, $\Omega \subseteq \mathcal{E}$, form an orthonormal basis. Their multiplication rule is

$$\chi_\Omega \chi_{\Omega'} = \chi_{\Omega \Delta \Omega'}.$$

We identify this basis with occupation vectors $|\Omega\rangle$; the *grade* is $|\Omega|$. Let $P_{\leq K}$ project onto grades at most K . The vertex support of an occupation state is the union of its edge endpoints, and has size at most $2|\Omega|$.

The multiplication rule is the whole point. Multiplying a Walsh character by a primitive sign W_a toggles a in and out of Ω , because $W_a^2 = 1$. Thus Ω records exactly the set of signs that have been multiplied in an odd number of times, and the expectation of a product of signs is 1 if the final state is $\Omega = \emptyset$ and 0 otherwise. We write $\emptyset$ for the empty set and call $|\emptyset\rangle$ the vacuum.

For a colored embedding ϕ , write $E_\phi = \{\{\phi(x), \phi(y)\} : xy \in E(H)\}$. Multiplication by its monomial sends

$$|\Omega\rangle \mapsto |\Omega \Delta E_\phi\rangle.$$

There are no signs in these occupation-matrix coefficients: they are nonnegative integers counting embedding multiplicities. This is the second benefit of the representation: all cancellation has been absorbed into the geometry of the basis, and the operators we must bound have nonnegative entries.

Let $\mathcal{H}_U, \mathcal{H}_V$ be the finite Euclidean spaces of colored boundary assignments. Multiplication by the self-adjoint dilation

$$\begin{pmatrix} 0 & M_H^{\text{col}} \\ (M_H^{\text{col}})^* & 0 \end{pmatrix}$$

has Walsh representation $\mathcal{W}$ on $(\mathcal{H}_U \oplus \mathcal{H}_V) \otimes \ell^2(2^\mathcal{E})$. Its direction from the U -register to the V -register is the adjoint matrix direction:

$$|a, \Omega\rangle \mapsto \sum_{\phi: \phi|_U = a} |\phi|_V, \Omega \Delta E_\phi\rangle. \quad (5.1)$$

The reverse block is its adjoint. Concretely, a vector in the U -register is a function of the row index a and the signs, expanded in Walsh characters; applying $(M_H^{\text{col}})^*$ to it produces a function of the column index $\phi|_V$, and each monomial of the matrix entry toggles its edges. We call the block (5.1) the *forward* block.

Example 5.1 (The single edge). Let H be the edge uv with $U = (u)$, $V = (v)$, and pools P_u, P_v . The forward block is

$$|a, \Omega\rangle \mapsto \sum_{b \in P_v} |b, \Omega \Delta \{a, b\}\rangle, \quad a \in P_u,$$

and the reverse block sends $|b, \Omega\rangle$ to $\sum_{a \in P_u} |a, \Omega \Delta \{a, b\}\rangle$. Starting from the vacuum, the forward block creates one edge $\{a, b\}$; to return to the vacuum, the reverse block must annihilate the same edge, which forces it to return to the same a . Hence $\langle a, \emptyset | \mathcal{W}^2 | a, \emptyset \rangle = |P_v|$, and summing over a gives $\mathbb{E} \text{Tr}(M_H^{\text{col}}(M_H^{\text{col}})^*) = |P_u||P_v|$, as it should. A closed walk of length $2q$ has, at each time, at most q edges created and not yet annihilated, which is the general fact proved next.

Lemma 5.2 (Vacuum compression). *For every integer $q \geq 1$, with $K = eq$,*

$$\begin{aligned} \mathbb{E} \text{Tr}((M_H^{\text{col}}(M_H^{\text{col}})^*)^q) &= \sum_a \langle a, \emptyset | (P_{\leq K} \mathcal{W} P_{\leq K})^{2q} | a, \emptyset \rangle \\ &\leq n^{|U|} \|P_{\leq K} \mathcal{W} P_{\leq K}\|^{2q}. \end{aligned} \quad (5.2)$$

Proof. In Walsh coordinates, expectation is the vacuum coefficient. The dilation's U -diagonal block after $2q$ steps is $(M_H^{\text{col}}(M_H^{\text{col}})^*)^q$, so the expected trace is the sum over a of the vacuum-to-vacuum matrix elements of $\mathcal{W}^{2q}$ starting at $|a, \emptyset\rangle$. Every step toggles e distinct primitive edges,

so it changes the grade by at most e in either direction. Any contributing walk that starts and ends at the vacuum has grade at time r at most

$$e \min(r, 2q - r) \leq eq,$$

since it needs at least as many steps to return to the vacuum as its current grade divided by e . Its entire walk is therefore retained by the compression, and the compressed and uncompressed matrix elements agree. There are at most $n^{|U|}$ legal U -assignments, and each compressed matrix element is at most the displayed operator norm to power $2q$. $\square$

The lemma reduces the moment problem to a deterministic one: bound the norm of $\mathcal{W}$ restricted to states of grade at most $K = eq$. We will do this uniformly over all occupied states of grade at most K , without using that the walk started at the vacuum. The dependence of the bound on K is then what determines the dependence of the moments on q , and hence the logarithmic factor.

5.2 Occupied patterns and their free labels

The forward block sums over all embeddings, whether or not their edges are already occupied. We now split it according to which edges are. For $F \subseteq E(H)$, let $\mathcal{A}_F$ be the forward block (5.1) restricted to embeddings satisfying

$$E_\phi \cap \Omega = \phi(F).$$

Then $\sum_F \mathcal{A}_F$ is the full forward block. In $\mathcal{A}_F$ the edges of F are annihilated and the edges outside F are created, so every $\mathcal{A}_F$ changes grade by the fixed amount $e - 2|F|$. We call F the *occupied pattern*.

Assume there are no isolated interior roles, and write $B = U \cap V$, $I = \mathcal{V} \setminus (U \cup V)$. Define

$$\begin{aligned} R_F &= \{x \in (U \setminus B) \cup I : \delta_H(x) \subseteq F\}, \\ Z_F &= \{x \in (V \setminus B) \cup I : \delta_H(x) \cap F = \emptyset\}, \quad S_F = \mathcal{V} \setminus (R_F \cup Z_F). \end{aligned} \quad (5.3)$$

Here $\delta_H(x)$ is the set of edges of H at x . An isolated exclusive boundary role belongs vacuously to the appropriate free set. Common boundary roles always stay in S_F . We call (R_F, Z_F, S_F) the *signature* of the pattern.

The names are chosen as follows. A role in Z_F is *fresh*: none of its edges is occupied, so in the forward direction its label ranges over its whole pool, subject only to the requirement that its newly created edges be unoccupied, which can only reduce the count. A role in R_F is *released*: all of its edges are annihilated, so in the reverse direction (where the roles of created and annihilated edges are exchanged) its label ranges over its whole pool in the same way. Every other role is *pinned*: either it is an input boundary role, or it has an edge in F , whose primitive image lies in Ω and whose endpoints therefore lie in the vertex support of Ω , a set of at most $2K$ labels. The next lemma turns this into a norm bound and identifies the combinatorial constraint on the free sets.

Example 5.3 (Signatures of the path). Let H be the path axb with $U = (a)$, $V = (b)$, so $h = 3$, $s = 1$, and $h - s = 2$. The four patterns have the following signatures.

F	R_F	Z_F	S_F	$ R_F + Z_F $
$\emptyset$	$\emptyset$	$\{x, b\}$	$\{a\}$	2
$\{ax\}$	$\{a\}$	$\{b\}$	$\{x\}$	2
$\{xb\}$	$\emptyset$	$\emptyset$	$\{a, x, b\}$	0
$\{ax, xb\}$	$\{a, x\}$	$\emptyset$	$\{b\}$	2

In the pattern $\{xb\}$ the edge xb is annihilated and ax is created: the label of x is pinned by the occupied edge, the label of b likewise, and a is the input. No label is free in either direction, and the pattern contributes at most a polynomial in K to the norm, against n for the other three. In each critical pattern S_F is one of the three minimum separators of the path.

Lemma 5.4 (Signature and Schur count). *The sets R_F, Z_F are disjoint, there is no edge between them, and S_F is a separator. Consequently*

$$|R_F| + |Z_F| \leq h - s.$$

At grades at most K , the forward column sums and reverse row sums of $\mathcal{A}_F$, including multiplicities, are respectively at most

$$n^{|Z_F|}(1 + 2K)^{h-|U|-|Z_F|}, \quad n^{|R_F|}(1 + 2K)^{h-|V|-|R_F|}. \quad (5.4)$$

If $|R_F| + |Z_F| < h - s$, then

$$\|P_{\leq K} \mathcal{A}_F P_{\leq K}\| \leq n^{(h-s-1)/2} (1 + 2K)^h. \quad (5.5)$$

Proof. Step 1: the signature is a separator. Only an isolated interior role could belong to both free sets, and such roles were excluded. An edge between the two sets would have to belong to F (because of its endpoint in R_F) and not belong to F (because of its endpoint in Z_F). Every surviving U -role lies in R_F , every surviving V -role in Z_F , where surviving means not in S_F ; hence a U -to- V path either starts or ends in S_F , or starts in R_F , ends in Z_F , and must cross from one to the other through S_F since there is no direct edge. Thus S_F separates the boundaries and its size is at least s , which gives $|R_F| + |Z_F| = h - |S_F| \leq h - s$.

Step 2: counting targets. Fix a forward source boundary assignment and occupied set Ω . Every noninput role incident to F must receive a label in the vertex support of Ω , which has at most $2K$ labels. The only unrestricted noninput roles are Z_F . This proves the first bound in (5.4). In the reverse direction the occupied embedding edges are precisely $E(H) \setminus F$, so its free roles are R_F , proving the second bound. Unused or inconsistent choices only overcount the embeddings.

Step 3: Schur. Apply Lemma 2.1. The resulting power of n is $(|R_F| + |Z_F|)/2$; strict inequality saves at least $n^{1/2}$. The remaining power of $1 + 2K$ is at most h . $\square$

We call F *critical* when equality holds, so S_F is a minimum separator. The noncritical patterns already have a strict polynomial saving, and (5.5) is all that will be used about them. Bounding the critical patterns is the next task, and it occupies Sections 6 and 7. The difficulty is that the Schur count of Lemma 5.4 gives only $n^{(h-s)/2} (1 + 2K)^h$ for a critical pattern, and the polynomial factor in K is far too large: at $K \asymp \log n$ it would produce $(\log n)^h$ instead of $(\log n)^{b_{\text{att}}(S_F)/2}$. The Schur test charges a factor $2K$ for every pinned label, but most pinned labels are not independent: they are determined by a few others through the edges of the shape. Recovering the correct power of K requires factoring the pattern at its separator and treating each piece with a sharper tool.

6 Creation estimates on arbitrary occupied states

This section proves the two operator estimates from which the critical bound of Section 7 is assembled. Both concern *pure creation* operators: operators that add the edges of an embedding to an occupied state, restricted to embeddings none of whose edges is already present. The first (Lemma 6.2) concerns a connected scalar shape with fixed root labels and is the source of the logarithm; the second (Lemma 6.4) concerns a shape whose boundary labels change and whose input boundary is a minimum separator, and it contributes no logarithm.

6.1 The bosonic analogy

The estimate to be proved has the form $\|\mathcal{C}|_{\text{grade } k}\| \leq \sqrt{k/e_J + 1} n^{c/2}(1 + o(1))$. The square root of the grade is familiar from the harmonic oscillator: if a^* is the boson creation operator, then $aa^* - a^*a = I$, and this relation alone implies $\|a^*|_{\text{grade } k}\|^2 = k + 1$, because on grade k one has $aa^* = a^*a + I$ and a^*a has norm $\|a^*|_{\text{grade } k-1}\|^2$, so the squared norms increase by exactly one per grade. The operator $\mathcal{C}$ below satisfies the same relation approximately, with n^c in place of I and with an error that is smaller by a factor n and polynomial in the grade. The following example shows the relation in its cleanest form.

Example 6.1 (The one-vertex field: an exact commutator). Let J have one root r with fixed label a , one interior role x with pool P , and the edge rx . The pure creation operator is

$$\mathcal{C}|\Omega\rangle = \sum_{j \in P: \{a,j\} \notin \Omega} |\Omega \cup \{a,j\}\rangle,$$

and it raises the grade by one. Let $N(\Omega)$ be the number of $j \in P$ with $\{a,j\} \in \Omega$. Applying $\mathcal{C}$ and then $\mathcal{C}^*$ adds an edge $\{a,j\}$ and removes an edge $\{a,j'\}$ of the result; if $j' = j$ the state returns to Ω , and there are $|P| - N(\Omega)$ such terms, while if $j' \neq j$ the state moves to $(\Omega \cup \{a,j\}) \setminus \{a,j'\}$ with $\{a,j'\} \in \Omega$ and $\{a,j\} \notin \Omega$. Applying $\mathcal{C}^*$ and then $\mathcal{C}$ removes $\{a,j'\} \in \Omega$ and adds $\{a,j\} \notin \Omega \setminus \{a,j'\}$; if $j = j'$ the state returns to Ω , with $N(\Omega)$ such terms, while if $j \neq j'$ the state moves to $(\Omega \setminus \{a,j'\}) \cup \{a,j\}$, the same state as before. The off-diagonal terms therefore cancel exactly, and

$$\mathcal{C}^*\mathcal{C} - \mathcal{C}\mathcal{C}^* = \text{diag}(|P| - 2N(\Omega)) \preceq |P|I.$$

Writing a_k for the squared norm of $\mathcal{C}$ on grade k , this gives $a_k \leq a_{k-1} + |P|$, hence $a_k \leq (k+1)|P|$ and $\|\mathcal{C}|_{\text{grade } k}\| \leq \sqrt{(k+1)|P|}$. This is the estimate of Lemma 6.2 with $c = 1$ and no error term: the error term of the lemma comes from pairs of embeddings that overlap partially, and a single vertex cannot overlap partially with another.

6.2 A connected scalar creation commutator

Take a rooted connected scalar shape J as in Section 4, fix its root labels, and let $e_J \geq 1$ be its number of edges after root-internal edges have been removed. Define pure creation by

$$\mathcal{C}|\Omega\rangle = \sum_{\phi: E_\phi \cap \Omega = \emptyset} |\Omega \cup E_\phi\rangle. \quad (6.1)$$

Let $\mathcal{C}_k$ be its block from grade k to grade $k + e_J$, with negative-grade blocks defined to be zero. Occupation states may contain arbitrary edges outside J 's edge-role blocks. This last point is important for the application: in the factorization of Section 7, the state on which a floating component acts also carries the edges of the sides and of the other components, and the estimate must hold regardless of them.

Lemma 6.2 (Connected creation commutator). *If c is the number of interior roles and $A_c = 2^c + c$, then*

$$\mathcal{C}_k^* \mathcal{C}_k - \mathcal{C}_{k-e_J} \mathcal{C}_{k-e_J}^* \preceq [n^c + A_c n^{c-1} (1 + 2k)^{c+1}] I. \quad (6.2)$$

Consequently

$$\begin{aligned} \|\mathcal{C}P_{\leq K}\|, \|\mathcal{C}^*P_{\leq K}\| &\leq \sqrt{K+1} n^{c/2} \\ &\quad + \sqrt{A_c} \sqrt{K+1} (1 + 2K)^{(c+1)/2} n^{(c-1)/2}. \end{aligned} \quad (6.3)$$

The left side of (6.2) is “add a copy of J , then remove a copy” minus “remove a copy, then add a copy,” both acting on grade k . The proof sorts the pairs of copies involved by how much they overlap. Disjoint pairs occur in both products with the same target and cancel exactly. Identical pairs return to the source state and give a diagonal term of size at most n^c . Partially overlapping pairs are the error term, and the estimate on them is where connectedness of the interior is used: if some but not all interior labels of the two copies differ, there is an edge of $J[C]$ from a changed role to an unchanged role, the unchanged endpoint carries a label that also belongs to an occupied edge, and that label is therefore one of the at most $2k$ labels in the support of Ω . One of the c free labels is thereby restricted, which costs a factor n against a polynomial in k .

Proof. Step 0: the two products. Fix a grade- k source Ω . A contribution to $\mathcal{C}_k^* \mathcal{C}_k$ adds an embedding ϕ and removes an embedding ψ :

$$E_\phi \cap \Omega = \emptyset, \quad E_\psi \subseteq \Omega \cup E_\phi, \quad \Omega' = (\Omega \cup E_\phi) \setminus E_\psi. \quad (+)$$

A contribution to $\mathcal{C}_{k-e_J} \mathcal{C}_{k-e_J}^*$ removes ψ and then adds ϕ :

$$E_\psi \subseteq \Omega, \quad E_\phi \cap (\Omega \setminus E_\psi) = \emptyset, \quad \Omega' = (\Omega \setminus E_\psi) \cup E_\phi. \quad (-)$$

In both cases the target Ω' is determined by the pair (ϕ, ψ) , so both products are sums over such pairs. Put $T = \{x \in C : \phi(x) \neq \psi(x)\}$, the set of changed interior roles. Role coloring makes agreement of interior image labels equivalent to agreement at the same role, so ϕ and ψ have disjoint interior images exactly when $T = C$, and coincide exactly when $T = \emptyset$.

Step 1: disjoint copies cancel. If the embeddings have disjoint interior image sets, then E_ϕ, E_ψ are disjoint, since every edge has an interior endpoint. Conditions (+) and (−) are equivalent in this case, since $E_\psi \subseteq \Omega \cup E_\phi$ with $E_\psi \cap E_\phi = \emptyset$ means $E_\psi \subseteq \Omega$, and $E_\phi \cap \Omega = \emptyset$ is then the same as $E_\phi \cap (\Omega \setminus E_\psi) = \emptyset$. They also give the same target. These terms cancel exactly, including their multiplicities.

Step 2: identical copies give the diagonal. If $T = \emptyset$, the embeddings coincide. They contribute a diagonal term equal to the number of addable embeddings minus the number of removable embeddings, and this is at most n^c , the total number of colored embeddings with the fixed root labels.

Step 3: partially overlapping copies, type (+). It remains to bound the absolute row sums of terms with $\emptyset \neq T \subsetneq C$. For a term of type (+), every ψ -edge incident to T differs from the corresponding ϕ -edge and cannot coincide with any other abstract edge role. It must therefore lie in Ω . Thus every changed ψ -label belongs to $\text{supp}_{\text{vert}} \Omega$. Connectedness supplies an interior edge from T to $C \setminus T$. At its unchanged endpoint y ,

$$\phi(y) = \psi(y) \in \text{supp}_{\text{vert}} \Omega,$$

because the edge of ψ at y towards T lies in Ω . Choose that endpoint canonically from T . One of the otherwise free ϕ -labels is restricted to at most $2k$ values, and the $|T|$ changed ψ -labels are restricted to at most $2k$ values each. For this fixed T there are at most

$$n^{c-1} (2k)^{|T|+1} \leq n^{c-1} (1 + 2k)^{c+1}$$

embedding pairs. Sum over at most 2^c choices of T .

Step 4: partially overlapping copies, type (−). For a term of type (−), all interior labels of the removed embedding ψ belong to $\text{supp}_{\text{vert}} \Omega$, because each has an incident edge, and all edges of ψ lie in Ω . There are at most $(2k)^c$ choices for ψ . The other embedding shares at least one

interior role, so choosing a shared role leaves at most cn^{c-1} choices for ϕ . This contributes at most $cn^{c-1}(1+2k)^c$. For either order the target state is determined by the embedding pair, so these are bounds on row sums with multiplicities.

Step 5: the operator inequality. After removing the diagonal terms, the remaining commutator is self-adjoint. Its absolute row sums, and hence its norm by Lemma 2.1, are at most $A_cn^{c-1}(1+2k)^{c+1}$, since the row-sum bounds of Steps 3 and 4 also bound column sums by the symmetry of the self-adjoint remainder. This proves (6.2). For $c = 1$ there is no nonempty proper T , and the same estimate is a valid enlargement.

Step 6: iteration along the grades. Let $a_k = \|\mathcal{C}_k\|^2$. The operator inequality implies

$$a_k \leq a_{k-e_J} + n^c + A_cn^{c-1}(1+2k)^{c+1},$$

since $\|\mathcal{C}_k\|^2 = \|\mathcal{C}_k^* \mathcal{C}_k\|$ and $\|\mathcal{C}_{k-e_J} \mathcal{C}_{k-e_J}^*\| = a_{k-e_J}$. Iterating in steps of e_J down to a negative grade, where the block is zero, gives

$$a_k \leq (\lfloor k/e_J \rfloor + 1)[n^c + A_cn^{c-1}(1+2k)^{c+1}].$$

Different grades have different target grades, so $\|\mathcal{C}P_{\leq K}\|^2 = \max_{k \leq K} a_k$. Taking the maximum over $k \leq K$, then using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, proves the first bound in (6.3). The adjoint on grade k is the adjoint of the creation block on grade $k - e_J$, proving the second. $\square$

Remark 6.3. The bound (6.3) has the form $\sqrt{K+1}n^{c/2}$ plus a term smaller by $n^{-1/2}$ and polynomial in K . At $K \asymp \log n$ the second term is negligible, and the first is $n^{c/2}(\log n)^{1/2}$ up to constants. Since a floating component of size c contributes $n^{c/2}$ to the polynomial order in any case, its entire effect on the logarithmic order is the factor $\sqrt{K+1}$, independently of c and of the number of attachment edges. This is why each attached floating component contributes exactly $1/2$ to the logarithmic exponent.

6.3 Creation from a minimum input boundary

We also need a side operator with changing boundary labels. Let L have input boundary S and output boundary O , possibly overlapping. Assume S is a minimum S -to- O separator in L , every component of $L - S$ meets O , and edges internal to S have been omitted. Suppose every role outside O has an incident edge. Isolated roles in O , including common boundary roles, are allowed. Let $v_L = |V(L)|$ and $\ell = v_L - |S|$. The pure-creation operator is

$$\mathcal{C}_L|a, \Omega\rangle = \sum_{\substack{\phi|_S=a \\ E_\phi \cap \Omega = \emptyset}} |\phi|_O, \Omega \cup E_\phi\rangle.$$

In the application, L will be the subgraph induced on a minimum separator S together with the components of $H - S$ meeting U , with $O = U$; the hypotheses are verified in Lemma 7.1.

The difference from the scalar case is that the boundary register changes, so there is no natural grading in which a commutator relation would be useful. Instead we bound the Gram matrix $\mathcal{C}_L^* \mathcal{C}_L$ on a single grade directly: its diagonal is at most n^ℓ , the number of ways to choose the ℓ labels outside S , and its off-diagonal entries are small for the same reason as in the scalar case, namely that two embeddings with the same output and the same target state but different inputs must overlap partially, which pins a label. The result is $\|\mathcal{C}_L\| \leq n^{\ell/2}(1+o(1))$ on every grade of polylogarithmic size, with no factor $\sqrt{K+1}$: the side operators do not contribute to the logarithm.

Lemma 6.4 (Side creation). *If $\ell \geq 1$, then*

$$\|\mathcal{C}_L P_{\leq K}\|, \|\mathcal{C}_L^* P_{\leq K}\| \leq n^{\ell/2} + 2^{v_L/2}(1+2K)^{v_L} n^{(\ell-1)/2}. \quad (6.4)$$

If $\ell = 0$, then $O = S = V(L)$, there are no edges, and $\mathcal{C}_L$ is the identity on the common boundary register.

Proof. Step 1: the degenerate case. The case $\ell = 0$ follows because $|O| \geq |S|$ by minimality, while $O \subseteq V(L) = S$. All edges are internal to S and have been omitted.

Step 2: the Gram diagonal. For $\ell \geq 1$, fix a grade- k source state (a, Ω) , and consider the Gram matrix $C_{L,k}^* C_{L,k}$. There are at most n^ℓ embeddings extending a . If two such embeddings have the same output assignment and target occupied set, their newly added edge supports agree. Every nonoutput role is determined by its incident edge, and every output role by the output assignment. Thus the embeddings agree. The Gram diagonal is at most n^ℓ .

Step 3: an off-diagonal term pins a frontier label. An off-diagonal term is a pair ϕ, ψ with the same output assignment and target occupation, but different source states. Let T be their nonempty changed-role set. It avoids O , and therefore avoids every common boundary role. Every ψ -edge incident to T differs from its ϕ -counterpart; role colors prevent a coincidence with a different edge role. Since the target occupied sets agree, it lies in the old set Ω . All changed ψ -labels consequently lie in its vertex support.

There must be a role

$$y \in (N_L(T) \setminus T) \setminus S. \quad (6.5)$$

Otherwise every neighbor outside T would lie in S . If $T \cap S = \emptyset$, the nonempty set T would be a union of components of $L - S$ missing O , a contradiction. If $T \cap S \neq \emptyset$, then $S \setminus T$ would separate the input S from O : the remaining input roles lie in T , which has no output role and cannot be left after the deletion. This contradicts minimality of S .

Step 4: the count. Choose y canonically from (6.5). It is unchanged, so $\phi(y) = \psi(y) \in \text{supp}_{\text{vert}} \Omega$, restricting one otherwise free ϕ -label to at most $2k$ choices. For a fixed T , the pair count is at most $n^{\ell-1} (2k)^{|T|+1}$. Given ϕ, ψ , the second source state is determined: its boundary labels are $\psi|_S$ and its occupied set is $(\Omega \cup E_\phi) \setminus E_\psi$. Thus the entire off-diagonal row sum is at most

$$2^{v_L} (1 + 2k)^{2v_L} n^{\ell-1}.$$

This also bounds column sums by symmetry. The assertion at $k = 0$ holds since no off-diagonal term can have the required old-support label. Lemma 2.1 bounds the Gram norm by this error plus n^ℓ .

Step 5: conclusion. Taking square roots proves the estimate on each grade. Different grades have distinct target grades, even if the number of edges is zero. Maximizing over $k \leq K$ proves the creation bound, and shifting the grade proves the adjoint bound. Isolated output roles are fixed in the Gram comparison and cause no extra multiplicity. $\square$

Remark 6.5 (Why the hypotheses are needed). The two structural hypotheses on L are used exactly once, in Step 3, to produce the frontier role y : a role adjacent to the changed set T , not in T , and not in S . Such a role is unchanged and has an occupied edge, so it pins a label. If some component of $L - S$ missed O , the changed set could be that whole component, with all its neighbors in S ; then no ϕ -label would be pinned, and the off-diagonal row sum would be comparable to the diagonal n^ℓ times a polynomial in the grade. This is exactly the situation of a floating component, where the norm does grow with the grade (Example 6.1). If S were not a minimum separator, the changed set could include a redundant separator role together with everything it cuts off, with the same effect. In the application both hypotheses come from the minimality of the separator of the critical pattern (Lemma 7.1), and they are the reason that the sides do not produce logarithms while the floating components do.

7 Critical separators and the core upper bound

In this section H is a core shape. By Lemma 3.1(i), assume there are no edges internal to either boundary. Boundary isolates and common roles remain allowed. Put $f_0 = (h - s)/2$.

The section has two parts. First, a critical pattern is shown to factor exactly, as an operator, into a product over the pieces of $H - S_F$: two side operators of the kind treated in Lemma 6.4 and one scalar creation or annihilation operator per floating component, of the kind treated in Lemma 6.2. Second, the resulting norm bound is inserted into the vacuum compression to obtain the core upper bound at logarithmic moments.

7.1 The exact side and floating factorization

Fix a critical occupied pattern F , and set $S = S_F$. By Lemma 5.4, S is a minimum separator. Every component of $H - S$ is entirely contained in R_F or entirely contained in Z_F , since there is no edge between those two sets. A component meeting U must lie in R_F ; one meeting V must lie in Z_F . Write $C_1, \dots, C_b$ for the remaining components. Since H is a core, each C_j has an edge to S , and $b = b_{\text{att}}(S)$.

Let L be the side graph on S and the components meeting U , with input S and output U . Let R be its counterpart for V . Include all edges incident to their vertices outside S ; omit edges internal to S . Put

$$\ell = |V(L) \setminus S|, \quad r = |V(R) \setminus S|, \quad c_j = |C_j|.$$

Then

$$\ell + r + \sum_{j=1}^b c_j = h - s, \tag{7.1}$$

since the vertices outside S are partitioned among the components meeting U , those meeting V , and the floating ones. This identity is where the polynomial exponent f_0 will come from: each piece contributes the square root of its own label count.

Lemma 7.1 (Side hypotheses). *The set S is a minimum input-to-output separator in both side graphs. Every component outside S in either side meets its output. Every role outside that output has positive degree. Thus Lemma 6.4 applies to both sides.*

Proof. The component statement follows from construction. If a smaller set separated U from S in the left side, it would separate U from V in the original graph: any original boundary-to-boundary path has a prefix ending at its first visit to S , and that prefix lies in the left side. A zero-length prefix is included when the path starts at a common role. This contradicts the minimality of S . The argument for the right side is the same, using the path in reverse.

A role in $S \setminus U$ with no left edge would be unnecessary in a separator between S and U , contradicting the just-proved minimality. A role outside $S \cup U$ in the left side belongs to a connected component meeting U , so it cannot be isolated. Only output roles can therefore be isolated. The right side is identical. $\square$

For each C_j , take its full root-role register to be S and include all its internal and attachment edges. Let $\mathcal{C}_j$ be its pure scalar creation operator, block diagonal in the separator assignment. Set

$$\mathcal{B}_j = \begin{cases} \mathcal{C}_j^*, & C_j \subseteq R_F, \\ \mathcal{C}_j, & C_j \subseteq Z_F. \end{cases}$$

Thus a floating component in the released set is annihilated and one in the fresh set is created; in both cases its labels are summed over freely. For a fixed separator assignment A , the edges internal to S have prescribed primitive images. Let $\mathcal{D}_{S, F \cap E(S)}$ project onto the occupancy pattern of these edges specified by F , and then toggle all of them. It preserves A and has norm at most one.

Lemma 7.2 (Exact critical operator product). *The forward occupation block has the exact factorization*

$$\mathcal{A}_F = \mathcal{C}_R \mathcal{D}_{S, F \cap E(S)} \left(\prod_{j=1}^b \mathcal{B}_j \right) \mathcal{C}_L^*. \quad (7.2)$$

The floating factors may be taken in any order. If source and target grades are at most K , every intermediate grade is at most $K + e$.

Read from right to left, (7.2) is an assembly line. The input is a U -assignment and an occupied state. The factor $\mathcal{C}_L^*$ annihilates the edges of the left side, which all lie in F , and in doing so reads off the separator labels, which become the new boundary register; the left labels outside S are summed out. Each floating factor then creates or annihilates the edges of its component, with the separator labels as roots, summing over the component's own labels. The separator factor tests and toggles the edges internal to S . Finally $\mathcal{C}_R$ creates the edges of the right side, none of which lies in F , choosing the right labels and producing the output V -assignment. The claim is that this sequential procedure enumerates exactly the embeddings of the pattern, each once.

Proof. Every edge incident to a left component belongs to F , so it is annihilated by $\mathcal{C}_L^*$. This factor chooses the left labels and the separator labels extending the input boundary. The latter labels are then stored in the intermediate register. Each floating factor chooses exactly its component labels and creates or annihilates its edges as required by the common signature of that component. The separator factor supplies the remaining tests and toggles. Every edge incident to a right component lies outside F , so $\mathcal{C}_R$ creates precisely those edges and chooses the remaining right labels.

The edge-role families of these factors are disjoint. By role coloring, their primitive edge families are disjoint as well. A factor therefore cannot alter another factor's occupancy tests: the test performed by one factor concerns only primitive edges in its own blocks, which no other factor touches. Distinct role pools also make all chosen labels globally injective. Consequently this procedure is a bijection, with multiplicities, between contributions to the product and embeddings satisfying $E_\phi \cap \Omega = \phi(F)$. It proves (7.2). The same disjointness permits reordering the floating factors.

At most the e edges of an embedding are changed during the product, so every intermediate grade is at most the source grade plus e . Larger grade projections can be inserted at every intermediate stage without changing a term with endpoint grades at most K . $\square$

Example 7.3 (The running example: the diagonal row-sum matrix). Let H be the edge rx with $U = V = (r)$, pools P_r and P_x , so that M_H^{col} is diagonal with entries $\sum_{j \in P_x} W_{aj}$, $a \in P_r$. The forward block is $|a, \Omega\rangle \mapsto \sum_{j \in P_x} |a, \Omega \triangle \{a, j\}\rangle$; it coincides with the reverse block. There are two patterns, $F = \emptyset$ and $F = \{rx\}$, both with signature $S_F = \{r\}$ and both critical, with the single floating component $\{x\}$ in $Z_\emptyset$ and in $R_{\{rx\}}$ respectively. Both sides are trivial ($\ell = r = 0$), there are no separator-internal edges, and (7.2) reads

$$\mathcal{A}_\emptyset = \mathcal{C}, \quad \mathcal{A}_{\{rx\}} = \mathcal{C}^*,$$

where $\mathcal{C}$ is the one-vertex creation operator of Example 6.1, block diagonal in a . That example gives $\|\mathcal{C}P_{\leq K}\|, \|\mathcal{C}^*P_{\leq K}\| \leq \sqrt{(K+1)n}$, so $\|P_{\leq K}\mathcal{W}P_{\leq K}\| \leq 2\sqrt{(K+1)n}$ with $K = q$, and Lemma 5.2 gives

$$\mathbb{E} \text{Tr}((M_H^{\text{col}}(M_H^{\text{col}})^*)^q) = \sum_{a \in P_r} \mathbb{E} \left(\sum_{j \in P_x} W_{aj} \right)^{2q} \leq n (2\sqrt{(q+1)n})^{2q}.$$

Taking $2q$ -th roots at $q = \lceil \log n \rceil$,

$$(\mathbb{E} \|M_H^{\text{col}}\|^{2q})^{1/(2q)} \leq n^{1/(2q)} \cdot 2\sqrt{q+1} \sqrt{n} \leq C\sqrt{n \log n}.$$

This is the upper bound of Theorem 1.3 for this shape, and it exhibits the source of the logarithm in the simplest case: the occupation operator is a creation operator plus its adjoint, its norm on grade k grows like $\sqrt{k+1}$, and the vacuum compression only needs grades up to $q \asymp \log n$.

Proposition 7.4 (Critical norm bound). *There are constants $C_H, D_H < \infty$ such that every critical pattern with separator S satisfies*

$$\|P_{\leq K} \mathcal{A}_F P_{\leq K}\| \leq C_H n^{f_0} (K+1)^{b_{\text{att}}(S)/2} + C_H n^{f_0-1/2} (K+1)^{D_H}. \quad (7.3)$$

The constants are uniform over all role pools, boundary assignments, and arbitrary occupied states. One may take $D_H = 6h + 4$.

Proof. Use the exact product and apply Lemma 6.4 to both sides, Lemma 6.2 to each floating factor, and the norm-one bound to $\mathcal{D}$, all at cutoff $K+e$. The norm of the product is at most the product of the norms of the factors, each restricted to grades at most $K+e$, which is legitimate by the last assertion of Lemma 7.2. Each factor's bound is a leading term plus an error term. The product of the leading terms is at most a fixed constant times

$$n^{\ell/2} n^{r/2} \prod_j (n^{c_j/2} \sqrt{K+e+1}) \leq C_H n^{f_0} (K+1)^{b/2},$$

by (7.1). Each other term in the finite product uses at least one error term and loses at least $n^{1/2}$. Its remaining dependence on $K+1$ is polynomial of fixed degree. For an explicit generous degree bound, the side exponents sum to $|V(L)| + |V(R)| \leq 2h$; each scalar exponent is bounded by $c_j + 2$, whose sum is at most $3h$. Thus $6h + 4$ exceeds every needed degree, including the noncritical bound used below. The shift by e changes only the shape-dependent constant. $\square$

Example 7.5 (The single edge, continued). For the edge uv with $U = (u)$, $V = (v)$ (Example 5.1), both patterns are critical with signatures $S_\emptyset = \{u\}$ and $S_{\{uv\}} = \{v\}$, and neither has a floating component. The pattern $\emptyset$ is the side creation operator $\mathcal{C}_R$ from $\{u\}$ to (v) , with $r = 1$, and the pattern $\{uv\}$ is $\mathcal{C}_L^*$ with $\ell = 1$. Lemma 6.4 gives $\|P_{\leq K} \mathcal{W} P_{\leq K}\| \leq 2n^{1/2} + O(K^2)$, and hence

$$(\mathbb{E} \|M_H^{\text{col}}\|^{2q})^{1/(2q)} \leq n^{1/(2q)} (2n^{1/2} + O(q^2)),$$

which is $O(n^{1/2})$ at $q \asymp \log n$: the order of a Wigner matrix, with no logarithm, as Theorem 1.3 predicts. Keeping track of the pool sizes in Lemma 6.4, the Gram diagonals are $|P_v|$ and $|P_u|$ rather than n , and letting q grow slightly faster than $\log n$ shows $\mathbb{E} \|M_H^{\text{col}}\| \leq (|P_u|^{1/2} + |P_v|^{1/2})(1 + o(1))$, which is the correct constant for the norm of a $|P_u| \times |P_v|$ matrix with independent sign entries. The machinery is therefore not wasteful on the simplest shape: no logarithm is produced where none is present.

7.2 From occupation norms to logarithmic upper moments

Proposition 7.6 (Core upper bound). *For every fixed core shape, every fixed $A > 0$, and arbitrary disjoint role pools of sizes at most n ,*

$$\| \|M_H^{\text{col}}\| \| \|_{L^p} \leq C_{H,A} n^{(h-s)/2} (\log(2n))^{b_*/2}, \quad 1 \leq p \leq A \log(2n). \quad (7.4)$$

The same upper bound holds for $M_{H,n}$.

Proof. There are 2^e forward patterns, and the reverse blocks are their adjoints. Apply (5.5) to noncritical patterns and (7.3) to critical patterns, then sum their norms. Because the number of patterns is fixed, this takes the largest floating count, not a sum of counts at different separators. After adjusting constants,

$$\|P_{\leq K} \mathcal{W} P_{\leq K}\| \leq C_H n^{f_0} (K+1)^{b_*/2} + C_H n^{f_0-1/2} (K+1)^{D_H}.$$

Vacuum compression gives the finite bound

$$\mathbb{E} \operatorname{Tr}((M_H^{\text{col}}(M_H^{\text{col}})^*)^q) \leq n^{|U|} \left[C_H n^{f_0} (q+1)^{b_*/2} + C_H n^{f_0-1/2} (q+1)^{D_H} \right]^{2q}. \quad (7.5)$$

Here $K = eq$, and e is absorbed into C_H .

Choose an integer $q \geq \max\{p/2, \log(2n)\}$ with $q \leq C_A \log(2n)$. For any rectangular matrix, $\|M\|^{2q} \leq \operatorname{Tr}((MM^*)^q)$, since the right side is the sum of the q -th powers of the eigenvalues of MM^* , all nonnegative, and the left side is the largest of them. The prefactor in (7.5) has bounded $2q$ -th root, because $n^{|U|/(2q)} \leq e^{|U|/2}$ when $q \geq \log(2n)$. Moreover,

$$n^{-1/2} (q+1)^{D_H-b_*/2} \rightarrow 0,$$

so the second term inside the bracket is eventually smaller than the first. Taking $2q$ -th roots and using monotonicity of moments ($p \leq 2q$) proves the colored estimate. The exact color average (Lemma 3.2) transfers it to the uncolored model. Boundary-internal edges removed at the start of this section are restored using Lemma 3.1(i).

Empty pools give a zero matrix unless the corresponding role set is empty. Zero-edge shapes also cause no exception: all roles are boundary roles, the scalar creation factors are absent, and the side operators are identities or all-ones output factors. The same signature and vacuum argument applies with grade zero. $\square$

The choice $q \asymp \log n$ is forced from both directions. It must be at least of order $\log n$ so that the prefactor $n^{|U|}$ disappears under the $2q$ -th root; and once q is of that order the factor $(q+1)^{b_*/2}$ becomes $(\log n)^{b_*/2}$, which the lower bound of the next section shows to be the truth. Larger q would give a weaker conclusion, since $(q+1)^{b_*/2}$ would then exceed the true order.

8 The matching core lower bound

This section proves the lower bound for core shapes. The strategy was described in Section 1.5: factor the colored matrix through the labels of a minimum separator, isolate the floating components as a diagonal factor, and test the matrix against the normalized columns of the side matrices at the separator label where the diagonal is largest.

Let H be a nonempty core shape, and choose a minimum separator S . Give every role a pool of size $m = \lfloor n/h \rfloor$. The side graphs L, R are constructed as in Section 7, but now use the ordinary random matrices, not occupation operators. We use L for the matrix with rows U and columns S , and R for the matrix with rows S and columns V . Edges internal to S are excluded from both. Let $C_1, \dots, C_b$ be the floating components at S . Explicitly, $L[a, A]$ is the sum over colored embeddings of the left side with $\phi|_U = a$ and $\phi|_S = A$ of the product of the left-side edge signs, and similarly for R .

Lemma 8.1 (Random-matrix separator product). *There is an exact factorization*

$$M_H^{\text{col}} = LDR, \quad D_{A,A} = \chi_{H[S]}(A) \prod_{j=1}^b F_{j,A}, \quad (8.1)$$

where $\chi_{H[S]}(A)$ is the product of signs on separator-internal edges. The random objects L, R, D are independent.

Proof. For fixed separator labels, the free roles of the two sides and the floating components are disjoint. Their prescribed role pools make injectivity automatic, so multiplying their sums and then summing over separator labels counts each embedding once: an embedding of H is the same thing as a separator assignment A together with embeddings of L , of R , and of each C_j extending A , and its monomial is the product of the monomials of the pieces. Common

boundary roles impose their equality constraints through the same separator-label register. This proves the identity. The three factors use disjoint primitive edge-role blocks, hence independent sets of signs. $\square$

This is the same decomposition as (7.2), now for the random matrix itself rather than for one occupied pattern of its occupation operator. For the lower bound only one separator is needed, and it is chosen to maximize b_{att} .

For a separator assignment A , define

$$u_A = \frac{Le_A}{\|Le_A\|}, \quad v_A = \frac{R^*e_A}{\|R^*e_A\|},$$

with either vector set to zero if its denominator vanishes. Thus u_A is the normalized A -th column of L and v_A the normalized A -th row of R , viewed as a column vector.

Lemma 8.2 (Normalized-column symmetry). *For each fixed A ,*

$$\mathbb{E}_L[L^*u_A] = e_A \mathbb{E}_L\|Le_A\|, \quad \mathbb{E}_R[Rv_A] = e_A \mathbb{E}_R\|R^*e_A\|. \quad (8.2)$$

Consequently

$$\mathbb{E}\|LDR\| \geq \alpha_L \alpha_R \mathbb{E} \max_A |D_{A,A}|, \quad \alpha_L = \mathbb{E}\|Le_A\|, \quad \alpha_R = \mathbb{E}\|R^*e_A\|, \quad (8.3)$$

where the latter expectations do not depend on A .

The idea is this. Since u_A and v_A are unit vectors, $\|LDR\| \geq |u_A^* LDR v_A|$, and

$$u_A^* LDR v_A = \sum_{B,B'} \langle u_A, Le_B \rangle D_{B,B'} \langle R^*e_{B'}, v_A \rangle = \sum_B \langle u_A, Le_B \rangle D_{B,B} \langle R^*e_B, v_A \rangle.$$

The term $B = A$ is $\|Le_A\| D_{A,A} \|R^*e_A\|$, which is what we want. The other terms involve the inner product of the column Le_B with the normalized column u_A , and the first identity in (8.2) says that this inner product has mean zero whenever $B \neq A$. The mechanism is a sign symmetry: there is a block of signs that enters every monomial of Le_B exactly once and does not enter Le_A at all, so flipping that block changes the sign of Le_B and leaves u_A unchanged. Such a block exists because S is minimal: a separator role z with $A_z \neq B_z$ has an edge into a left component, and the primitive signs on that edge role with z -label B_z form the block.

Proof. To prove the first identity, take $B \neq A$ and choose a separator role z with $B_z \neq A_z$. If $z \in U$, the columns Le_A and Le_B have disjoint row support: the common boundary coordinate is fixed to different labels. Their inner product is then zero.

Otherwise $z \notin U$. Minimality of S implies that $S \setminus \{z\}$ fails to separate U, V . A witnessing path, stopped at its first separator vertex, reaches z through an edge zx with x in a left component. Flip every primitive sign in this one abstract edge block whose z -label is B_z . Every monomial in column Le_B changes sign exactly once, since each colored embedding with $\phi(z) = B_z$ uses exactly one primitive sign from the block $\mathcal{B}_{zx}$ with z -label B_z . Column Le_A and its norm are unchanged, since $A_z \neq B_z$ and different abstract edge blocks cannot coincide. This measure-preserving sign flip gives $\mathbb{E}_L \langle Le_B, u_A \rangle = 0$. The A -coordinate is exactly $\mathbb{E}_L \|Le_A\|$, including when the column is zero. This proves the first identity in (8.2). The right identity follows in the same way with the right side.

Choose $A = A(D)$ as a maximizer of $|D_{A,A}|$, with fixed tie-breaking. Conditional on D , the test vectors depend separately on L and R . Independence and (8.2) yield

$$\mathbb{E}_{L,R}[u_A^* LDR v_A \mid D] = D_{A,A} \alpha_L \alpha_R.$$

Within-pool relabeling makes the two column expectations independent of the particular tuple A . Pointwise, $\text{sgn}(D_{A,A}) u_A^* LDR v_A \leq \|LDR\|$, since the test vectors have norms at most one. Take conditional and then full expectations to obtain (8.3). $\square$

Lemma 8.3 (Column scale). *If $\ell = |V(L) \setminus S|$ and $r = |V(R) \setminus S|$, then*

$$\mathbb{E}\|Le_A\|^2 = m^\ell, \quad \mathbb{E}\|R^*e_A\|^2 = m^r,$$

and

$$\alpha_L \geq c_H m^{\ell/2}, \quad \alpha_R \geq c_H m^{r/2}. \quad (8.4)$$

Proof. Expand the squared column norm. In a fixed output coordinate, a pair of monomials survives only if its two primitive edge supports agree: both monomials are square-free products of signs. Every summed interior role has an incident edge and is identified by its color, so these supports, together with the common output labels, identify the entire embedding. Isolated output roles merely duplicate entries over their free coordinates. There are exactly m^ℓ complete free assignments, each contributing one. This proves the left second moment; the right proof is identical.

Each column entry is a sign polynomial of degree bounded by e . Lemma 2.3 then supplies (8.4): the column is a vector of fixed-degree sign polynomials, so its expected norm is at least a constant times its root-mean-square norm. When a side has no roles outside S , its register is an identity and its column norm is one. $\square$

Proposition 8.4 (Core lower bound). *For every fixed core shape,*

$$\mathbb{E}\|M_{H,n}\| \geq c_H n^{(h-s)/2} (\log(2n))^{b_*/2}$$

for all sufficiently large n .

Proof. The empty shape is immediate. Otherwise choose a minimum separator attaining b_* and use equal pools of size $\lfloor n/h \rfloor$. Every floating component attaches to S , so Lemma 4.5 implies

$$\mathbb{E} \max_A |D_{A,A}| \geq c_H m^{(\sum_j c_j)/2} (\log m)^{b_*/2},$$

since the event in (4.9) has probability at least $1/2$ eventually and $|\chi_{H[S]}(A)| = 1$. For $b_* = 0$, the maximum is one and the same assertion holds. Combining Lemmas 8.2 and 8.3, and $\ell + r + \sum_j c_j = h - s$, gives

$$\mathbb{E}\|M_H^{\text{col}}\| \geq c_H m^{(h-s)/2} (\log m)^{b_*/2}.$$

The Fourier comparison in Lemma 3.3 applies because a core has no isolated interiors. Since $m \asymp_H n$, it gives the required uncolored bound. $\square$

Example 8.5 (The running example: the lower bound). For the edge rx with $U = V = (r)$, the separator is $S = \{r\}$, both sides are empty ($\ell = r = 0$, and $L = R = I$), and the single floating component $\{x\}$ gives $D_{a,a} = F_a = \sum_{j \in P_x} W_{aj}$. Thus $M_H^{\text{col}} = D$ and $\|M_H^{\text{col}}\| = \max_a |F_a|$ exactly, with no need for the normalized-column argument. The fields F_a for distinct $a \in P_r$ use disjoint sign arrays and have no interior signs, so they are independent and Lemma 4.5 is just the statement that the maximum of m independent sums of m signs is at least $c\sqrt{m \log m}$ with probability tending to one. Lemma 3.3 with $a_H = 1$ then returns to the uncolored diagonal matrix. The path axb with $U = V = (a, b)$ is similar, with D indexed by pairs and $F_{(i,j)} = \sum_x W_{ix} W_{xj}$. The first shape in which the sides are nontrivial is, for instance, the path uzx with $U = (u)$ and $V = (z)$ and a pendant x at z : there $S = \{z\}$, the left side is the edge uz with $\ell = 1$, the right side is trivial, $D_{A,A} = \sum_j W_{Aj}$, and Lemma 8.2 is what allows the maximum over A to be extracted from under the left factor.

Remark 8.6. The lower bound proves more than the expectation bound: the colored event in Lemma 4.5 has probability tending to one. What does *not* follow is a high-probability statement for the uncolored matrix, because Lemma 3.3 compares moments, not realizations. Section 9 upgrades the expectation lower bound to a positive-probability statement by a second-moment argument, which is what is needed for all fixed L^p sizes.

9 Disconnected components and completion of the theorem

This section removes the two simplifying assumptions made so far. A general shape may have original connected components that miss both boundaries, and it may have isolated interior vertices. The first kind of component is shown to contribute an independent scalar factor whose L^p size is $n^{c/2}\sqrt{p}$ and whose expected absolute value is of order $n^{c/2}$; the second contributes an explicit deterministic factor. The section ends by extending the two-sided bound from expectations to all fixed positive L^p sizes.

9.1 Original boundary-free components

Assume first that H has no isolated interior vertices. Remove its original boundary-free connected components $G_1, \dots, G_t$ to obtain the possibly empty core H_0 . Each G_ν has at least one edge; write $c_\nu = |V(G_\nu)|$ and $h_0 = |V(H_0)|$. By Lemma 3.1, s, b_* are those of the core.

For every fixed coloring there is the exact algebraic identity

$$M_H^{\text{col}} = M_{H_0}^{\text{col}} \prod_{\nu=1}^t X_{G_\nu}^{\text{col}}, \quad (9.1)$$

where $X_{G_\nu}^{\text{col}}$ is the rootless colored scalar sum of that component. The identity holds because a colored embedding of H is the same thing as a colored embedding of H_0 together with one of each G_ν , with injectivity guaranteed by the disjoint pools, and its monomial is the product of the monomials of the pieces. Under full independence all these factors are independent, since their edge-role blocks are disjoint.

Upper bound. For arbitrary pools of sizes at most n , Corollary 4.3 gives

$$\|X_{G_\nu}^{\text{col}}\|_{L^p} \leq C_{H,A} n^{c_\nu/2} \sqrt{p}, \quad 1 \leq p \leq A \log(2n).$$

Together with Proposition 7.6 and (9.1), using independence to factor the expectation of the product of the p -th powers, this proves

$$\| \| M_H^{\text{col}} \| \|_{L^p} \leq C_{H,A} n^{(h-s)/2} (\log(2n))^{b_*/2} p^{t/2}. \quad (9.2)$$

The empty core contributes the scalar one. Color averaging proves the same bound for the uncolored matrix.

Lower bound. For the lower bound, assign equal pools of size $m = \lfloor n/h \rfloor$ to every role of the entire shape. Orthogonality gives $\mathbb{E}|X_{G_\nu}^{\text{col}}|^2 = m^{c_\nu}$: distinct colored embeddings have distinct sign supports, so the cross terms vanish, and there are m^{c_ν} embeddings. Its degree is fixed, so Lemma 2.3 gives

$$\mathbb{E}|X_{G_\nu}^{\text{col}}| \geq c_H m^{c_\nu/2}.$$

Independence in (9.1) and the colored core lower proof give

$$\mathbb{E} \| M_H^{\text{col}} \| = \mathbb{E} \| M_{H_0}^{\text{col}} \| \prod_{\nu} \mathbb{E} |X_{G_\nu}^{\text{col}}| \geq c_H m^{(h-s)/2} (\log m)^{b_*/2}.$$

Lemma 3.3 again transfers this to the uncolored matrix. In particular, no original boundary-free component incurs a logarithmic cost in expectation.

The contrast with an attached floating component is instructive. An attached component is indexed by the separator labels, and the operator norm selects the label at which it is largest, which costs $(\log n)^{1/2}$. An original boundary-free component is a single scalar, not indexed by anything; its typical size is $n^{c/2}$, and that is what the expectation sees. Its large deviations are visible only in higher moments, which is the $p^{t/2}$ in (9.2).

Remark 9.1. The factorization (9.1) is asserted after coloring. The corresponding uncolored matrix need not factor: global injectivity couples its component labels, as in the sixth row of Example 1.2, where the scalar factor excludes the labels used by the edge. The exact color average and Fourier comparison are what justify returning to the original model.

9.2 Isolated interiors and fixed positive moments

Now delete the i_0 isolated interior roles to obtain H' . For $n \geq h$, the exact factor in (3.1) is asymptotic to n^{i_0} , and s, b_* do not change. The polynomial exponent therefore becomes

$$\frac{|V(H')| - s}{2} + i_0 = \frac{h + i_0 - s}{2}.$$

We have proved the expectation lower bound and the strong upper bound (1.5) under full independence for every shape.

Let $Z = \|M_{H,n}\|$ and $T = T_{H,n}$. The estimates at $p = 1, 2$ give

$$\mathbb{E}Z \geq c_H T, \quad \mathbb{E}Z^2 \leq C_H T^2.$$

Lemma 2.2 applied to Z gives constants $c, d > 0$, depending only on the shape, such that

$$\mathbb{P}\{Z \geq cT\} \geq d$$

eventually. This proves $\|Z\|_{L^p} \geq cd^{1/p}T$ for every fixed $p > 0$. For fixed $p \geq 1$, the upper bound follows from (1.5); for $0 < p < 1$, monotonicity gives $\|Z\|_{L^p} \leq \mathbb{E}Z$. This completes the proof of Theorem 1.3.

10 Upper tails and limited independence

This section draws two consequences of the fact that the upper bound was proved at moments of logarithmic order and used only expectations of polynomials of logarithmic degree in the signs.

10.1 Polynomial upper tails

Fix $D > 0$, and choose $p = a \log(2n)$ for a fixed $a > 0$. Theorem 1.4 under full independence implies

$$\| \|M_{H,n}\| \|_{L^p} \leq C_{H,a} a^{t/2} T_{H,n} (\log(2n))^{t/2}.$$

Markov's inequality at a sufficiently large fixed multiple of this scale gives

$$\mathbb{P}\{\|M_{H,n}\| > C_{H,D} T_{H,n} (\log(2n))^{t/2}\} \leq n^{-D} \quad (10.1)$$

eventually: the probability that $\|M_{H,n}\|$ exceeds e times its L^p norm is at most $e^{-p} = (2n)^{-a}$, and a may be taken larger than D . For $t = 0$, this is the expected scale itself. Taking $D > 1$ and summing the failure probabilities shows that the corresponding upper bound holds eventually almost surely under *any* coupling of the matrices across sizes. No independence between different sizes is needed.

For $t > 0$, a fixed-threshold vanishing upper-tail statement at $T_{H,n}$ cannot be inferred from the expectation classification. For instance, the rootless single-edge scalar is $2 \sum_{i < j} W_{ij}$, whose polynomial normalization retains ordinary sum fluctuations: it exceeds any fixed multiple of its standard deviation $\asymp n$ with probability bounded away from zero. The additional $p^{t/2}$ in the strong-moment statement records this distinction.

10.2 An upper-only finite-independence transfer

Unbiased k -wise independent signs have exactly the same expectations as independent signs for every polynomial of total degree at most k : reduce powers using $W_a^2 = 1$, and apply independence to each remaining square-free monomial, which involves at most k signs. The upper proof only ever computes such expectations, so it transfers.

First consider a fixed colored core. The trace polynomial

$$\text{Tr}((M_H^{\text{col}}(M_H^{\text{col}})^*)^q)$$

has degree at most $2e_H q$. If $2e_H q \leq k_n$, its expectation matches the fully independent one, so (7.5) transfers exactly. Taking q a sufficiently large fixed multiple of $\log(2n)$, with $2q \geq p$, proves the core upper bound uniformly for any prescribed $p \leq A' \log(2n)$, provided $k_n \geq K_{H,A'} \log(2n)$. If there are no edges the assertion is deterministic.

For the full shape without isolated interiors, retain the algebraic colored factorization (9.1). Its factors may now be dependent. Hölder nevertheless gives

$$\| \| M_H^{\text{col}} \| \|_{L^p} \leq \| \| M_{H_0}^{\text{col}} \| \|_{L^{(t+1)p}} \prod_{\nu=1}^t \| X_{G_\nu}^{\text{col}} \|_{L^{(t+1)p}}. \quad (10.2)$$

Transfer the core upper bound through $(t+1)p \leq (t+1)A \log(2n)$. For a scalar factor choose an even order $2r$ with $(t+1)p \leq 2r < (t+1)p + 2$. The polynomial $(X_{G_\nu}^{\text{col}})^{2r}$ has degree at most $2r|E(G_\nu)|$, so its moment also transfers under a sufficiently large logarithmic independence constant. The independent scalar estimate bounds its $L^{(t+1)p}$ size by $C_H n^{c_\nu/2} \sqrt{p}$. Insert these bounds in (10.2).

Finally use the exact color average, which is an algebraic identity for every sign realization, and the deterministic isolated-interior factor. This proves the limited-independence part of Theorem 1.4, including its $p^{t/2}$ dependence.

The lower proof does not transfer by this argument. Its conditional root trials and the Fourier contraction use more than matching a single low-degree moment: the excursion argument conditions on some signs and uses the independence of the others, and the Fourier projection uses the invariance of the law under block sign flips. No matching logarithmic lower assertion for arbitrary logarithmic-wise independent signs is made here.

11 Consequences, examples, and scope

11.1 Exact no-log criterion

Since b_* is a nonnegative integer, the logarithmic exponent is always a nonnegative half-integer. For a core shape,

$$g = 0 \quad \Longleftrightarrow \quad \text{every component of } H - S \text{ meets a boundary for every } S \in \mathcal{S}_{\min}.$$

For a general shape, apply this condition to its boundary-containing core. Original boundary-free components and isolated interiors change the polynomial scaling but not this criterion.

The number of vertices within one connected floating component affects the polynomial exponent. Its number of attachments does not alter its logarithmic increment: as soon as at least one attachment exists, that connected component contributes $1/2$ at that separator. This is visible in the proof: the floating factor in (7.2) is the creation operator of the whole component, and Lemma 6.2 charges it $\sqrt{K} + 1$ once, however many edges attach it.

The upper sum ranges over finitely many critical patterns, whereas the lower proof chooses one maximizing separator. This is why the formula uses the maximum over separators, not a sum. A non-minimum separator S' with more floating components does not compete: the pattern with signature S' is bounded by $n^{(h-|S'|)/2}$ times a polynomial in K , and $n^{(|S'|-s)/2}$ exceeds any power of $\log n$.

11.2 Examples

Example 11.1 (An ordinary edge and parallel paths). For an edge with opposite singleton boundaries, $h = 2, s = 1, b_* = 0$, so the expected norm has order $n^{1/2}$. For $k \geq 1$ parallel length-two paths between singleton terminals, $h = k + 2, s = 1, b_* = 0$. Indeed deleting any minimum separator leaves no boundary-free component: deleting a terminal leaves one component containing the other terminal, and for $k = 1$ deleting the middle vertex leaves the two terminals. The expected order is $n^{(k+1)/2}$.

Example 11.2 (A common boundary can create an extreme). For the edge rx with $U = V = (r)$, the unique minimum separator is $\{r\}$. The remaining singleton x is attached, so $f = 1/2, g = 1/2$. The matrix is diagonal with row-sum entries $\sum_{j \neq i} W_{ij}$, and its expected norm has order $\sqrt{n \log n}$; this is the running example of Sections 6–8. For the wedge $a - x - b$ with $U = V = (a, b)$, the unique minimum separator is $\{a, b\}$, again giving $f = g = 1/2$.

Example 11.3 (Crossed parallel interiors). Let $U = (a, b)$, $V = (c, d)$, include edges ac, bd , and add $k \geq 1$ distinct interiors x_j with edges ax_j, x_jd . The disjoint edges ac, bd force $s \geq 2$, since each is a U -to- V path and no single vertex meets both, and $\{a, d\}$ is a separator of size two: every path from $\{a, b\}$ to $\{c, d\}$ either starts at a , or starts at b , whose only neighbor is d . It leaves k attached interior singletons, since each x_j is adjacent only to a and d . No separator can have more floating components than the k interior vertices. Hence

$$f = 1 + \frac{k}{2}, \quad g = \frac{k}{2}, \quad \mathbb{E}\|M_H\| = \Theta_H(n^{1+k/2}(\log n)^{k/2}).$$

This example shows that the logarithmic exponent can be arbitrarily large, and that the floating components of a single separator all contribute.

Example 11.4 (Competing endpoint branches). Take an edge uv , $U = (u), V = (v)$, with k_L arbitrary connected interior branches attached only to u and k_R attached only to v . Each branch has at least one attachment edge and there are no edges between branches. If the total number of interior roles is T , then the only minimum separators are $\{u\}, \{v\}$. Their floating counts are k_L, k_R , respectively. Thus

$$\mathbb{E}\|M_H\| = \Theta_H\left(n^{(T+1)/2}(\log n)^{\max(k_L, k_R)/2}\right).$$

This example shows that the maximum in (1.2) is not a sum: the branches at u and the branches at v cannot be extreme simultaneously, because they are floating at different separators.

Example 11.5 (Global components and isolated interiors). For a connected rootless graph on $c \geq 2$ vertices, both boundaries are empty and the unique minimum separator is empty. Thus $s = 0, b_* = 0$, and the expected absolute scalar value has order $n^{c/2}$. For i isolated interior roles and empty boundaries the scalar is exactly $(n)_i$, giving polynomial exponent i , as predicted by the isolated-interior correction.

11.3 What the theorem does not assert

The proof concerns fixed simple shapes and symmetric dense sign variables. It does not establish a classification for sparse inputs, Gaussian or unbounded inputs, multigraphs, growing shapes, or noninjective homomorphism matrices. It gives no general sharp leading constant or complete spectral law.

The uncolored lower bound is an expectation bound, strengthened by a second-moment argument to a positive fixed probability and to every fixed positive L^p size. The proof does not make that uncolored probability tend to one. The high-probability simultaneous excursions in Lemma 4.5 belong to the colored construction, and the Fourier moment comparison does not turn them into a pointwise comparison.

The following restrictions are part of the statements.

- The logarithmic lower bound uses full independence, through the conditional trials of Lemma 4.5 and the sign-law invariance in Lemma 3.3. The upper bounds transfer to logarithmically-wise independent signs; the lower bound is not claimed to.
- The constants $c_{H,p}, C_{H,p}$ are not explicit. The dominant losses in the upper bound are the factor h^h from Lemma 3.2, the factor 2^e from the number of patterns, and the shift of the grade cutoff by e ; in the lower bound they are the constant c_H of Lemma 4.5, which depends on b_* through the choice of δ , and the fourth-moment constant c_e of Lemma 2.3.
- The shape is fixed and $n \rightarrow \infty$; no uniformity in h is claimed, and the threshold $n_0(H)$ is not tracked.

A Self-contained dependencies and verification record

The proof uses only finite-dimensional linear algebra, elementary inequalities on probability spaces, and the lemmas proved here. In particular:

- The fixed-degree fourth-moment estimate is proved in Lemma 2.3.
- The scalar moment estimate and conditional concentration are proved by canonical equality-pattern counts in Propositions 4.2 and 4.4.
- The occupation representation and its grade cutoff are exact finite identities, established in Section 5.
- The critical operator estimate is proved on arbitrary occupied states, not inferred from scalar vacuum moments.
- The colored-to-uncolored lower transfer is proved by explicit conditional expectation and signed Fourier averaging.

No statement in the main proof requires access to the earlier project reports.

Dependency table. The following table records, for each main result, the statements its proof invokes. Reading it downward gives the logical order of the document; reading it upward from Theorem 1.3 gives the minimal set of statements needed for each half of the theorem.

Result	Depends on
Lemma 2.3 (fourth moments)	Lemma 2.2
Lemma 3.2 (color average)	—
Lemma 3.3 (Fourier projection)	—
Proposition 4.2 (scalar moments)	—
Corollary 4.3	Proposition 4.2
Proposition 4.4 (two-replica count)	Proposition 4.2
Lemma 4.5 (excursions)	Lemma 2.2, Proposition 4.4
Lemma 5.2 (vacuum compression)	—
Lemma 5.4 (signatures)	Lemma 2.1
Lemma 6.2 (creation commutator)	Lemma 2.1
Lemma 6.4 (side creation)	Lemma 2.1
Lemma 7.2 (critical product)	Lemma 5.4, Lemma 7.1
Proposition 7.4 (critical bound)	Lemmas 7.2, 6.2, 6.4
Proposition 7.6 (core upper)	Lemmas 5.2, 5.4, 3.2, 3.1; Proposition 7.4
Lemma 8.2 (normalized columns)	—
Lemma 8.3 (column scale)	Lemma 2.3

Result	Depends on
Proposition 8.4 (core lower)	Lemmas 8.1, 8.2, 8.3, 4.5, 3.3
Theorem 1.3	Propositions 7.6, 8.4; Corollary 4.3; Lemmas 3.1, 2.3, 2.2
Theorem 1.4	Proposition 7.6; Corollary 4.3; Lemma 3.2; Section 10

Verification record. The fixed-positive-moment classification corresponding to Theorem 1.3, its expected-norm specialization, and the nonnegative half-integer logarithmic exponent have been formalized in Lean 4.33.0 with pinned Mathlib dependencies. On 8 September 2026, [verification workflow 7](#) passed for [source commit 0199436](#). The run completed the Lean build and transitive axiom audit, Comparator’s comparison of the independent Challenge and Solution statements and their fixed definition dependencies, independent NanoDa proof replay, and Lean kernel replay. The permitted axioms were `propext`, `Quot.sound`, and `Classical.choice`.

This certification applies to those formal theorem statements and their proof dependencies in the accompanying Lean development. It does not extend to every intermediate statement or textual derivation in this manuscript. In particular, Theorem 1.4, including its uniform logarithmic moment range and asymptotic limited-independence assertion, and the upper-tail and almost-sure extensions are not formalized in full.

The source arguments also underwent internal mathematical review before this manuscript was assembled. Supplementary finite checks covered all 16,933 graph/boundary configurations on at most four vertices and 37 exact colored occupation factorizations, including extraneous occupied edges. These finite checks supplement the proofs and the separate formal verification; they are not themselves asymptotic proofs or formal proof-assistant certificates. External human peer review is not claimed, and the recorded checks do not establish publication priority or independent discovery. The mathematical examples and proof guide point to the technical arguments or to explicit finite computations. The historical and theorem-level comparisons instead refer to the cited sources, in particular the other proof in [4, Theorem 1.3]. The attribution revision does not replay the formal artifact.

B Notation index

Symbol	Meaning
<i>Shape and model</i>	
$H = (\mathcal{V}, E), h, e$	The shape graph, its number of roles, and its number of edges.
$U, V, B = U \cap V, I$	Ordered boundary lists (also used as sets), the common boundary, and the interior $\mathcal{V} \setminus (U \cup V)$.
W_{ij}	Independent uniform signs, $W_{ji} = W_{ij}$, $W_{ii} = 0$; a <i>primitive sign</i> or <i>primitive edge</i> .
$M_{H,n}$	The graph matrix (1.1).
ϕ, E_ϕ	An injective embedding and its set of primitive edges $\{\{\phi(x), \phi(y)\} : xy \in E\}$.
$s, \mathcal{S}_{\min}$	Minimum separator size and the family of minimum separators (Definition 1.1).
$b_{\text{att}}(S), b_*$	Attached floating components at S and their maximum over $\mathcal{S}_{\min}$, (1.2).
i_0, t	Number of isolated interior roles; number of original boundary-free components with an edge.
$f, g, T_{H,n}$	Exponents and scale, (1.3).
$\ X\ _{L^p}, (m)_q, d_q$	$(\mathbb{E} X ^p)^{1/p}$; falling factorial; $(2q - 1)!!$.

Symbol	Meaning
<i>Coloring</i>	
P_x, M_H^{col}	Role pools and the colored matrix.
$\mathcal{B}_{xy}$	The block of primitive edges available to the edge role xy .
$a_H, \mathcal{Q}$	Boundary-fixing automorphism count and the signed block average (3.4).
<i>Scalar fields</i>	
J, R, C, c	A rooted connected scalar shape, its roots, its interior, and $ C $.
F_A	The colored field (4.1) at root labels A .
$B_c, \rho, N_{q,d}$	$6c + 8$; the ratio $(2cq)^{B_c}/n$; the number of equality patterns of defect d .
$\omega, x, Z_q(\omega), \mu_q$	Interior signs, root-to-interior signs, the conditional $2q$ -th moment, and its mean (Section 4.2).
$X_{G_\nu}^{\text{col}}$	Rootless scalar of an original boundary-free component, (9.1).
<i>Occupation operators</i>	
$\mathcal{E}, \Omega, \Omega\rangle, \emptyset$	All label pairs; an occupied set; its Walsh vector; the vacuum $\Omega = \emptyset$.
grade, $P_{\leq K}, K = eq$	$ \Omega $; projection onto grades at most K ; the cutoff in Lemma 5.2.
$\text{supp}_{\text{vert}} \Omega$	Vertex support of Ω , of size at most $2 \Omega $.
$\mathcal{W}, \mathcal{A}_F$	Walsh representation of the dilation; its forward block with occupied pattern F .
R_F, Z_F, S_F	Released, fresh, and pinned roles of a pattern, (5.3).
$\mathcal{C}, \mathcal{C}_k, \mathcal{C}_L$	Pure creation of a rooted scalar shape (6.1), its grade- k block, and the side creation operator.
A_c, e_J	$2^c + c$ and the edge count of J after removing root-internal edges.
L, R, ℓ, r, C_j, c_j	Side graphs at a separator, their numbers of roles outside S , the floating components, and their sizes.
$\mathcal{B}_j, \mathcal{D}_{S,F \cap E(S)}$	Floating factors and the separator factor in (7.2).
f_0, D_H	$(h - s)/2$ and the polynomial degree $6h + 4$ in (7.3).
<i>Lower bound</i>	
m	Common pool size $\lfloor n/h \rfloor$.
L, D, R	The random-matrix factors of (8.1).
$u_A, v_A, \alpha_L, \alpha_R$	Normalized column and row test vectors; expected column norms (8.3).
$\chi_{H[S]}(A)$	Product of the separator-internal signs at A .

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